# EIGSEP Memo 008

Marjum Pass 2026-07: image-domain modeling of the sky, the ground and the Sun

EIGSEP Team

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**Abstract**

How calibrated Marjum spectra are modelled in the image domain: as the beam-weighted sum of a Galactic sky map, a terrain at one uniform temperature below the DEM horizon, a receiver offset, and the Sun. The memo covers the model and the degeneracies its geometry leaves, the night fit to the calibrated 07-17/18 data, and the Sun. Box-air records solar radio bursts at the times independent spectrographs recorded them, which identifies solar emission in the data. The quiet Sun is harder: the ridges hide it for hours, so it appears only at a handful of terrain crossings. The one calibrated crossing, 15:54 UTC on 07-17, shows an excess 3.7 times the concurrent Sun's prediction (1.5 scaled by the template amplitude), while the raw data at the other crossings allow a Sun of the expected size but do not detect it convincingly. Each measurement is shown against the raw data around it. Finally, regressions on the rotating antenna give a repeatable ground term, too frequency-dependent to be the ground's temperature alone. Every number, figure and table is computed by `make_products.py` in this directory from the products under `derived/ground_sky/` (Appendix A). The kelvin scale is memo 004's, the beam is HFSS (memo 005 measures it), and pointing is memo 003's.

## 1 Summary

- **The model** is linear at each frequency: sky pixels, one ground temperature, one offset per receiver regime and, optionally, the Sun's flux density. The columns are integrated through an assumed beam on the real geometry (Section 2). A uniform shift of sky and ground against the offset is an exact null direction.
- **The DEM stops short of the southern horizon.** Extending it 4 km south raises the horizon by up to 5.47° at 87.5 m and changes the predicted antenna temperature by up to 47 K, with the sky and beam fixed (Section 2, Fig. 1). The extended DEM and its horizon are now `derived/dem/v0004` and `horizon_profiles_v0004`; the fits here still use v0003.
- **The ground temperature is not measured by the calibrated data.** They hold one pointing, one height and one regime at night (165 two-minute bins), where $T_{\rm gnd}$ and the offset are degenerate. Only their sum, the pedestal, is determined: 8 to 91 K at 110–170 MHz. A forecast on the full campaign geometry shows that height changes are what would break the degeneracy (Table 1).
- **The data vary about twice as much as GSM through HFSS predicts**: the template amplitude is $a \approx 2.1$ (1.3–3.0 over the band). The model also does not reach the noise: median $\chi^2$ per bin is 120, and 234 on held-out blocks.
- **When the Sun is visible.** The ridges stand 39° high at 352 m to the east and 17° at 171 m to the west. The Sun clears the ridge at **15:54 UTC on 07-17** (87.5 m), 3.5 h after it reaches 0° altitude, and sets behind the ridge at **01:21 on 07-18** (91 m). The calibrated data therefore see the direct Sun only from 15:54 to 16:22 on 07-17. At these wavelengths the edge diffracts, so the Sun fades in over 10–30 min.
- **Box-air sees the Sun's radio bursts.** At the times e-CALLISTO spectrographs recorded type III bursts, raw box-air shows emission drifting from high to low frequency within seconds, bright at 45–85 MHz as those stations saw it. 5 of 12 bursts with data pass a test that 5.2% of random daytime times pass ($p = 2.2 \times 10^{-4}$). That identifies solar emission in our data physically (§6.5).
- **The quiet Sun across days.** All 6 terrain crossings of phase C are shown in their raw context; 3 of them, on three days and three heights, can be fitted. They allow a Sun term of 3–16 SFU at 121–219 MHz, the size of the concurrent Sun (RSTN at 245 MHz, steady at 13–17 SFU from 07-15 to 07-18). They do

not detect it convincingly, since raw power drifts by as much for other reasons (§6.7). None shows a jump as large as the calibrated 07-17 one.

- **The calibrated 07-17 sunrise excess is larger than the Sun.** After 15:54 the calibrated spectrum rises by a smooth, additive 20–25 K at 148–199 MHz (Fig. 5). That is 3.7 times what concurrent RSTN flux predicts (1.5 times scaled by $a$), and 2.0–3.2 times the other crossings at 199 MHz. A local fit gives $S_\odot$ from 7 SFU at 109 MHz to 32 SFU at 199 MHz; the same fit at the 0° sunrise, with the Sun behind the ridge, gives −6 to 1 SFU. Part of the excess is probably not the Sun. Something that changes as the box comes into sunlight would follow the same geometry.
- **Impulsive events in daytime are local**, not solar: their rate is the same with the Sun behind the ridge as with it visible (§6.6).
- **Not yet possible**: separating the Sun from the lit ground by timing, or from the sky and ground by the beam's sweep during the raster (§6.8). Below 90 MHz the calibrated result is unusable.
- **The ground from the rotating antenna.** When the antenna moves, the ground fraction swings while the gain stays put, so the ground/GSM ratio of a gain-free regression gives the ground's brightness in GSM kelvin. Three 91 m windows agree on it. It falls from 496 K at 125 MHz and 338 K at 139 MHz to 192–205 K at 150–175 MHz and reaches thousands below 85 MHz, which a reflected-sky term does not explain (§7). A joint fit across frequency and the sky-calibrated parked nights do not separate the ground, and low-order corrections to GSM are not separable from ground-fixed structure.

## 2 The model

At each frequency, every time bin $t$ is modelled as

$$T(t) = \sum_p A_{\rm sky}[t,p]\, T_{\rm sky}[p] + A_{\rm gnd}[t]\, T_{\rm gnd} + T_{\rm off}[r(t)] + A_\odot[t]\, S_\odot, \tag{1}$$

with the sky in Galactic HEALPix pixels ($N_{\rm side} = 8$), the ground everything below the DEM horizon at one temperature, an offset per receiver regime $r$, and the Sun's flux density $S_\odot$ in SFU ($10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$). The sky is written $T_{\rm sky} = a \cdot \mathrm{GSM} + \delta$. The amplitude $a$ is free; it absorbs the front-end loss (calibrated temperatures are at reference plane P, memo 004) and any GSM scale error. Only $\delta$ has a prior, a Gaussian of 10% of GSM.

**Integration.** The sky and ground columns are integrated on a topocentric grid ($N_{\rm side} = 64$). For each distinct pointing and height era, the beam (HFSS total power here) is rotated by the mount, `mount_rotation`$(\mathrm{az} + \mathrm{az}_0, \mathrm{el} + \mathrm{el}_0, \psi)$, with $\mathrm{az}_0 = -16.04°$, $\mathrm{el}_0 = +0.93°$ and $\psi = 142.2°$ from the reference beam fit `empirical_raster_v0013` (memo 005), the frame in which its empirical beam was fitted. It is cut at that era's DEM horizon (`horizon_profiles_v0003`, traced from the antenna of geometry release `v0004`, memo 002, on the DEM mosaic `derived/dem/v0003`). Only the map from grid directions to Galactic pixels changes with time. Rows are normalized by the full-sphere beam integral, so sky and ground weights sum to one. That makes a uniform shift of sky and ground, against the offset, an exact null direction: breaking it needs a known offset or a known sky. The solver (`eigsep_sim.design_matrix.solve`) reports which columns move along a null direction and marks them unconstrained.

**Bearings.** The DEM, and so the horizon profiles, are in NAD83(2011)/UTM 12N. Grid north there is 1.52° from true north. The fit rotates the horizon to true bearings; the first trials of 2026-10-01 did not.

**Terrain coverage.** The horizon profiles are traced out to the edge of the DEM mosaic `derived/dem/v0003`, which reaches $N = -1000$ m on its southern side. To see what that edge costs, we extended the mosaic 4 km south on the same grid, keeping every original pixel, the origin, the resolution and the vertical datum, and traced the horizon again from the profile antenna position $(E, N, U) = (1656.2, 2031.8, 1778.6)$ m (the 91 m era of geometry release `v0004`) in the DEM frame. The 30, 87.5 and 91 m height labels subtract 61, 3.5 and 0 m from $U$. Bearings are true bearings, rotated from grid north by 1.52° as above, in 0.1° steps with 720 sub-samples of the mask. Only the DEM changes between the two traces: sky, beam, site, times, pointings and frequencies are the same. The southern terrain raises the horizon by up to 3.45°, 5.47° and 5.51° at 30, 87.5 and 91 m. At the

eight observed night pointings and four frequencies (51–246 MHz), the predicted antenna temperature changes by at most 16, 47 and 48 K (Fig. 1). The 87.5 m case is the selected night (07-17, 04:22–11:23 UTC); the 30 and 91 m cases are synthetic height controls evaluated at those same times, not observations from those eras. These are differences between two model predictions, not measured errors in any calibrated spectrum. A third trace with 8 km of added terrain differs from the 4 km one by at most $4.1 \times 10^{-12}$ K, at the nominal position and six $\pm 3$ mm displacements. Edge flags alone do not test coverage: the original DEM raised none at 30 m and still changed. The tracing is also not shown to be accurate to sub-mK: the finest bearing refinement tested changes the prediction by up to 7.6 mK, so the numbers above are diagnostic and not tied to any convergence bound. The fits in this memo use the original DEM and have not been rerun on the extended one.

The extended terrain is now a product. `derived/dem/v0004` is v0003 plus 6 km to the south, on the same grid and keeping every v0003 pixel, and `curation/horizon_profiles_v0004` is the horizon `build_horizon_profiles.py` traces on it. Over v0003 it rises by at most 3.45, 5.47 and 5.51° at 30, 87.5 and 91 m, on 209 bearings at 87.5 m, and is unchanged elsewhere. Terrain outside v0004 cannot change it: `coverage_certificate.py` follows every bearing beyond the DEM edge on the USGS 1 arc-second DEM, out to the distance (at most 42 km) beyond which terrain below 4200 m would sit under the skyline, and finds none that rises above it, from the profile position at all three heights or from the eight displaced positions of the profile's sensitivity block (`curation/horizon_profiles_v0004_coverage.json`). The coarse terrain is raised by 8.5 m, the most by which it falls below the skylines traced inside the DEM; the smallest margin is then 0.99°, on southern bearings, and 2.57° in the other directions. The check covers this antenna position only.

**The Sun column.** The Sun is a point source:

$$A_\odot(t) = \frac{\lambda^2}{8\pi k}\, G(\hat{s}(t))\, V(t) \quad \text{(K per SFU)}, \tag{2}$$

for one linear polarization and an unpolarized source. Here $G$ is the beam gain toward the Sun, with the same full-sphere normalization as the rows, and $V$ is its terrain visibility. $V$ is not read off the fixed horizon profile. The DEM is traced along the Sun's own bearing at each time, which gives the edge's elevation and its distance $d$. The ridges are 150–350 m away, so at these wavelengths the edge diffracts over $\sqrt{\lambda d/2} \approx 2$–$6°$, tens of minutes of solar motion. $V$ is the Fresnel knife-edge power for the Sun's angle $\theta$ above the edge, $v = -\theta\sqrt{2d/\lambda}$: it is 0.25 at $\theta = 0$ and rises to 1 with lit-side ripples. The real ridge is rough and rounded, which smooths these ripples.

The knife-edge shape is an assumption. No measurement here tests it: its frequency-dependent onset, its value at the edge, its ripples. Every fit uses $A_\odot$ only as a template, with one fitted amplitude and, in the timing scans, one time shift. Figure 4 shows it against the data it was fitted to.

# 3 What the geometry can constrain

Before any data are used, `degeneracy_study.py` builds the design matrix on the campaign's real pointing and heights. It reports Fisher errors on $T_{\rm gnd}$ with radiometer noise in 120 s bins (Table 1).

- With the sky free, $T_{\rm gnd}$ is never constrained. That holds even with a known offset: within one pointing and era the ground column is constant and trades against a uniform sky shift.
- With a 10% GSM prior and the offset free, $T_{\rm gnd}$ and the offset are anticorrelated at $\rho = -0.95$ to $-1.00$. Height changes, which move the horizon, break the degeneracy: all of phase C would give 0.8–16 K.
- The calibrated windows alone are weak unless the offset is known.

The study also simulates and recovers. With HFSS as the true beam the recovered $T_{\rm gnd}$ is unbiased. With the empirical beam as truth and HFSS assumed, it is biased by up to 39 K ($16\sigma$) at 78 MHz, while $\chi^2$ per row moves only from 3.1 to 3.5. A good $\chi^2$ cannot vouch for $T_{\rm gnd}$.

# 4 Data

Box-air, antenna-switch rows calibrated by `tcal@v0003` (memo 004), which cover 07-17 04:11–16:22 (87.5 m, receiver regime rx-A) and 07-18 01:27–02:56 (91 m, rx-B). Samples flagged by `flags@v3` (memo 006) are dropped,

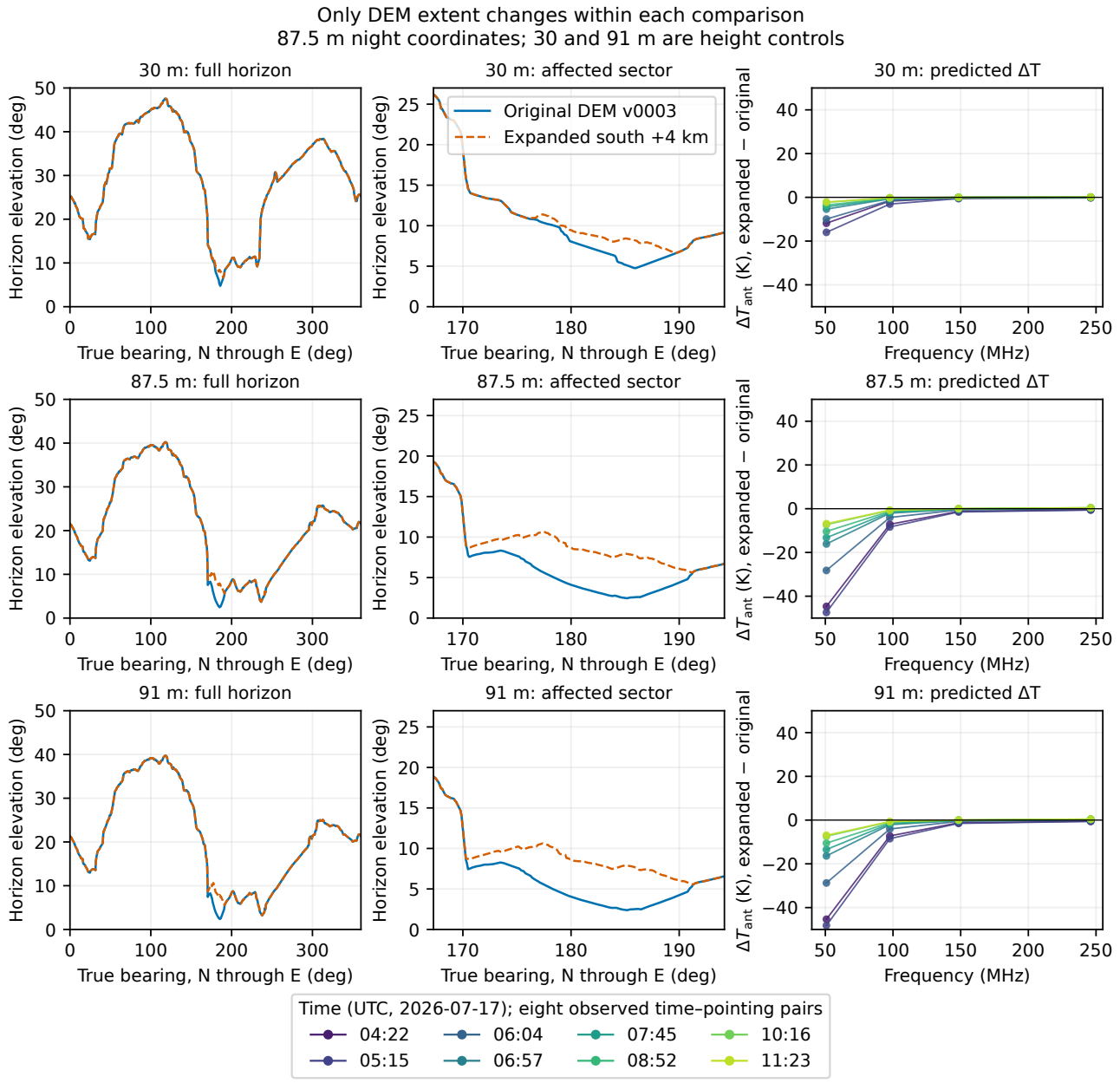



Figure 1: Effect of extending the DEM 4 km south, at the nominal antenna position, for the three height labels (rows). The horizon from the original DEM (`derived/dem/v0003`) and the extended DEM, with the affected southern sector enlarged, and the extended-minus-original predicted temperature at four frequencies, with the sky and beam fixed. Colours mark the eight observed night time/pointing pairs by UTC on 07-17; points are the evaluations and lines guide the eye. Within each pair only the DEM changes. The 30 and 91 m rows are synthetic height controls at the same times. The model azimuth is 0.865° for all eight pairs and the model elevations are 0.63–1.33° (`inputs/terrain_coverage/coordinates.csv`). Bearings are true; the horizon step is 0.1°.

| slice | 58.6 | 78.1 | 101.6 | 148.4 | 199.2 |
|---|---|---|---|---|---|
| *10% GSM sky prior, offset free* | | | | | |
| parked, 30 m only | 100.4 | 47.8 | 26.6 | 15.9 | 17.9 |
| parked, all eras | 21.4 | 10.0 | 5.4 | 2.8 | 1.1 |
| all of phase C | 16.0 | 7.4 | 3.9 | 1.7 | 0.8 |
| tcal windows only | 186.7 | 46.3 | 13.6 | 86.8 | 2.3 |
| *10% GSM sky prior, offset known to 1 K* | | | | | |
| parked, 30 m only | 7.9 | 3.7 | 2.2 | 2.0 | 1.7 |
| parked, all eras | 5.0 | 2.5 | 1.6 | 1.4 | 0.8 |
| all of phase C | 3.3 | 1.7 | 1.2 | 1.0 | 0.6 |
| tcal windows only | 11.4 | 5.4 | 2.9 | 2.2 | 1.2 |

Table 1: Forecast $\sigma(T_{\rm gnd})$ in K at five frequencies (MHz), from the degeneracy study (`summary.json` in `data-analysis`; radiometer noise, 120 s bins). "–": unconstrained. This forecast predates `tcal` v0003 and uses the v0000 calibrated windows.

on any bit but the advisory bit 9. From 07-17 15:36 the transmitter's comb channels (every eighth channel) are dropped whatever the flags say. At each HFSS beam frequency the remaining channels within ±2 are averaged. Rows are then averaged into static bins of one era, regime and pointing (az and el each within 1°): 302 two-minute bins (275 rx-A, 27 rx-B; product `binned_v0001`) and 615 one-minute bins (`binned_v0002`). Each bin's noise is the standard error of its rows. Frequencies whose in-bin scatter exceeds ten times the radiometer expectation are not fitted, which removes FM and leaves 44.

# 5 The night fit

`fit_v0009` fits Eq. 1 without the Sun, to the 165 bins with the Sun below −10° (Fig. 2, Table 2). All of them are rx-A, at one pointing and one height.

- $T_{\rm gnd}$ **is not measured.** It is exactly degenerate with the offset, as Section 3 predicts. At 24 frequencies the ±1° elevation wobble formally separates them. The values there run from −6890 to 14655 K: fitted model error, not a temperature. The pedestal $\langle f_{\rm gnd}\rangle T_{\rm gnd} + T_{\rm off}$ is determined: 8 to 91 K at 110–170 MHz.
- **The sky amplitude is** $a \approx 2.1$, measured to ±0.02 at each frequency. Two explanations remain open: the `tcal` kelvin scale (it assumes the nameplate noise-source excess, memo 004), or a sky fraction that HFSS underestimates. The fit cannot tell them apart.
- **The model is not adequate at the noise level**: median $\chi^2$ per bin is 120, and 234 on held-out alternating 30-min blocks.
- **The morning behind the ridge looks like night.** Adding the bins from 0° sunrise to the ridge crossing, `fit_v0010` (264 bins), raises the median $\chi^2$ only from 120 to 136.

| product | bins fitted | bins | freqs | median $\chi^2$ | median held-out $\chi^2$ |
|---|---|---|---|---|---|
| `fit_v0009` | night (Sun below −10°) | 165 / 302 | 44 | 120 | 234 |
| `fit_v0010` | Sun behind the ridge allowed | 264 / 302 | 44 | 136 | 256 |
| `fit_v0011` | night, 60 s bins | 319 / 615 | 44 | 111 | 242 |
| `fit_v0012` | Sun modelled, all bins | 302 / 302 | 44 | 486 | 55394 |

Table 2: The fit products. Every fit evaluates the model on all bins; "bins fitted" are those used to solve. `fit_v0012` fits the Sun column on every bin, including those with the transmitter on; its held-out $\chi^2$ reaches $3 \times 10^6$, and it is not used.

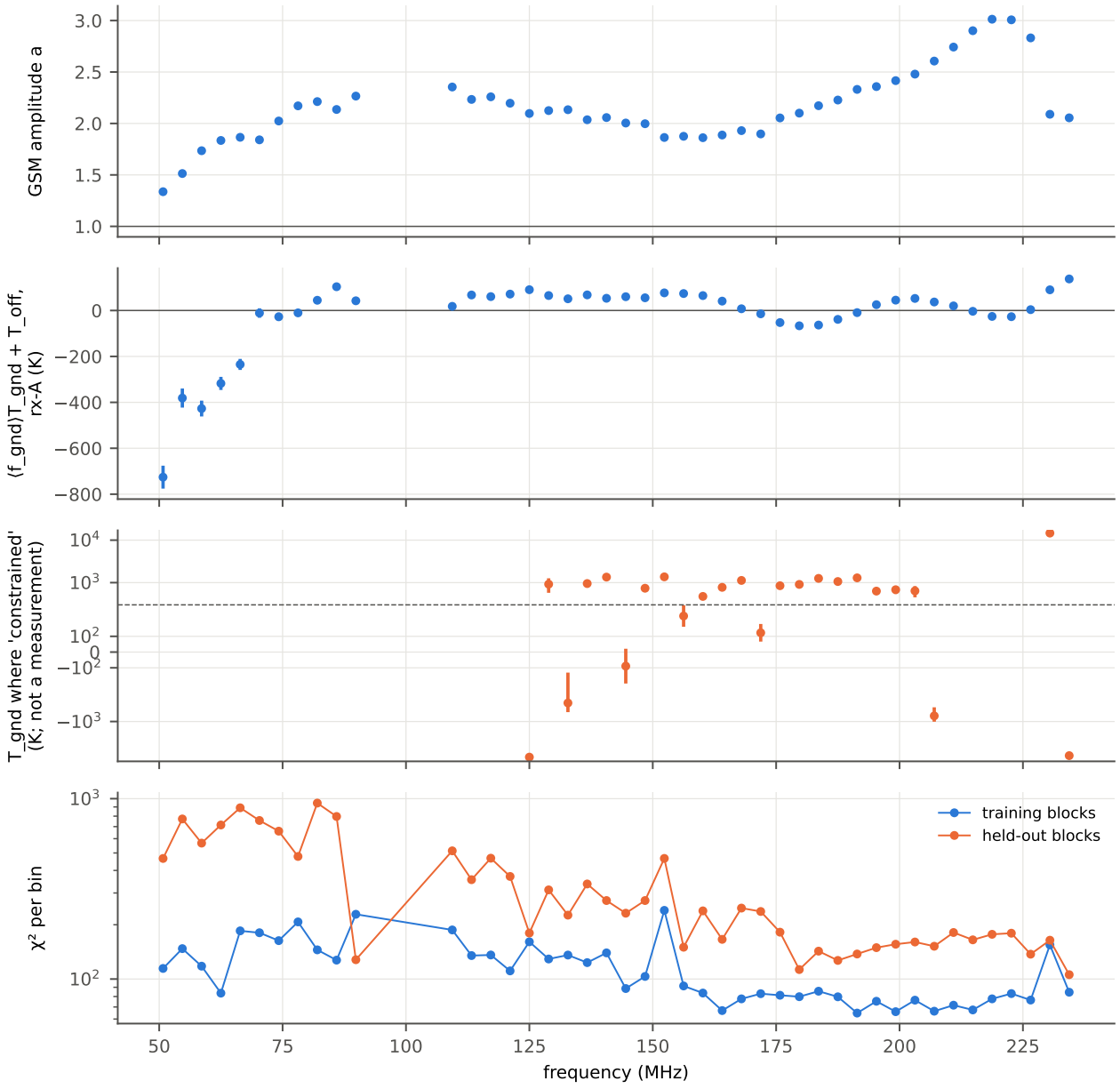



Figure 2: The night fit (`fit_v0009`). From the top: GSM amplitude; the rx-A pedestal; $T_{\mathrm{gnd}}$ where the solver calls it constrained (symlog axis; the dashed line is 300 K); $\chi^2$ per bin on training and held-out blocks.

# 6 The Sun

## 6.1 When it is visible

Table 3 lists every time the Sun's centre crosses the DEM horizon or 0° altitude while each height era was flown in phase C. Figure 3 shows the horizons and the Sun's tracks. At 87.5 m the Sun clears the eastern ridge at 15:54 UTC on 07-17, at 39° altitude and azimuth 94°, with the edge 352 m away. That is 3.5 h after it reaches 0° altitude at 12:26. At 91 m it sets behind the western ridge at 01:21 on 07-18 (17°, edge 171 m), 1.5 h before 0° (02:53).

Two consequences follow. The rx-B window starts at 01:27, after the terrain sunset, so of the calibrated data only 07-17 15:54–16:22 sees the direct Sun. And the 2026-10-01 trials' observation that the Sun "lifts 199 MHz within an hour of rising" cannot be the direct Sun: that hour is behind the ridge.

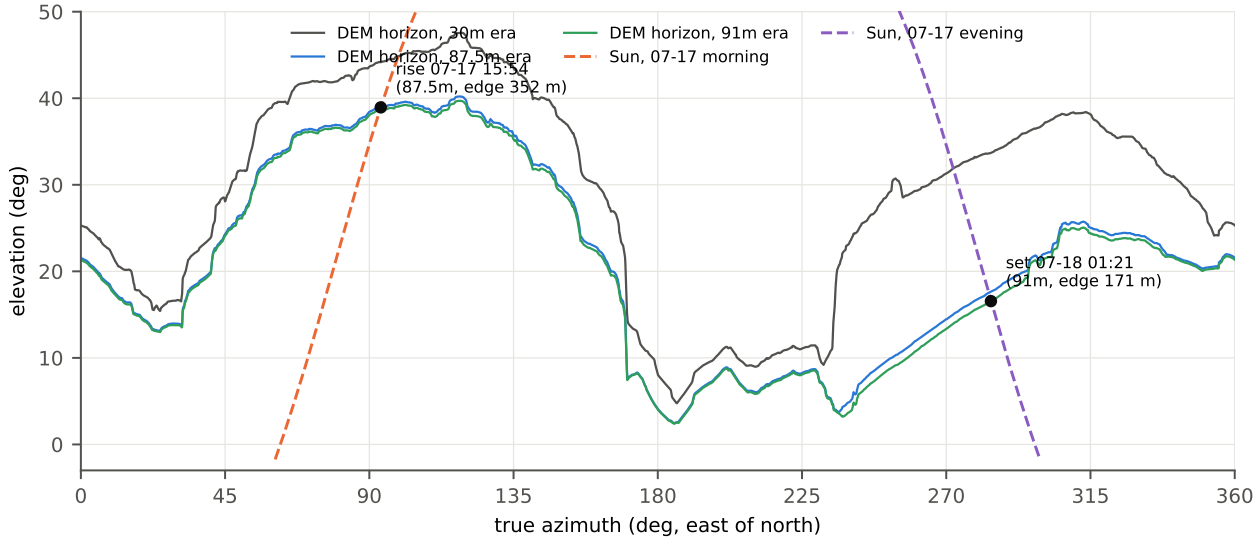



Figure 3: DEM horizon against true azimuth for the three heights (`horizon_profiles_v0003`, rotated from grid to true bearings), and the Sun's tracks on 07-17. Dots mark the terrain crossings in the calibrated eras.

| era | event | horizon | UTC | Sun alt (deg) | Sun az (deg) | horizon (deg) | edge (m) |
|---|---|---|---|---|---|---|---|
| 30m | set | flat | 07-15 02:54:57 | -0.0 | 298.3 | 0 | – |
| 30m | rise | flat | 07-15 12:24:47 | 0.0 | 61.8 | 0 | – |
| 30m | rise | terrain | 07-15 16:23:57 | 44.8 | 98.4 | 44.8 | 335 |
| 30m | set | terrain | 07-16 00:01:47 | 32.0 | 272.6 | 32.0 | 151 |
| 30m | set | flat | 07-16 02:54:17 | -0.0 | 298.1 | 0 | – |
| 30m | rise | flat | 07-16 12:25:27 | 0.0 | 62.0 | 0 | – |
| 30m | rise | terrain | 07-16 16:24:17 | 44.8 | 98.7 | 44.8 | 335 |
| 30m | set | terrain | 07-17 00:01:37 | 31.9 | 272.4 | 31.9 | 151 |
| 87.5m | set | flat | 07-17 02:53:44 | -0.0 | 297.9 | 0 | – |
| 87.5m | rise | flat | 07-17 12:26:14 | 0.0 | 62.3 | 0 | – |
| 87.5m | rise | terrain | 07-17 15:54:34 | 38.9 | 93.6 | 38.9 | 352 |
| 91m | set | terrain | 07-18 01:21:30 | 16.6 | 283.9 | 16.6 | 171 |
| 91m | set | flat | 07-18 02:53:10 | -0.0 | 297.6 | 0 | – |

Table 3: Sunrises and sunsets during phase C, by height era (`sun_events_v0003`). "terrain": the Sun's centre crosses the DEM horizon traced along its bearing; "flat": 0° altitude. Geometric, with no refraction.

## 6.2 The test around 15:54

The test reads `fit_v0011`, a night fit on one-minute bins. Every bin with the Sun above 0° is out of sample there, so its residual (data minus fitted sky, ground and offset) is a prediction. That residual drifts by tens of kelvin over hours, as the night model is carried into the morning. So each frequency is fitted locally, from 14:30 to 16:23 UTC, as

$$\mathrm{resid}(t) = c_0 + c_1(t - t_{\mathrm{ridge}}) + S_\odot\, A_\odot(t - \tau).$$

The transmitter of the beam-mapping raster is on 15:36–16:14, with short gaps (box-gnd transitions, memo 001). Its bins are never fitted. That leaves 62 bins, of which only 7 (16:14–16:22) have the Sun above the ridge. Errors on $S_\odot$ are scaled by $\sqrt{\chi^2/\mathrm{dof}}$.

Each of the following checks could have come out the other way:

1. **A null at 0° sunrise.** The same fit around 12:26 (97 transmitter-off bins), with a Sun column that ignores the terrain, should give $S_\odot \approx 0$ if the ridge hides the Sun.
2. **Timing.** $\chi^2$ against a shift $\tau$ of the Sun column should be smallest near $\tau = 0$.
3. **Spectrum.** $S_\odot(\nu)$ should be positive and smooth, of the order of the quiet-to-active Sun.

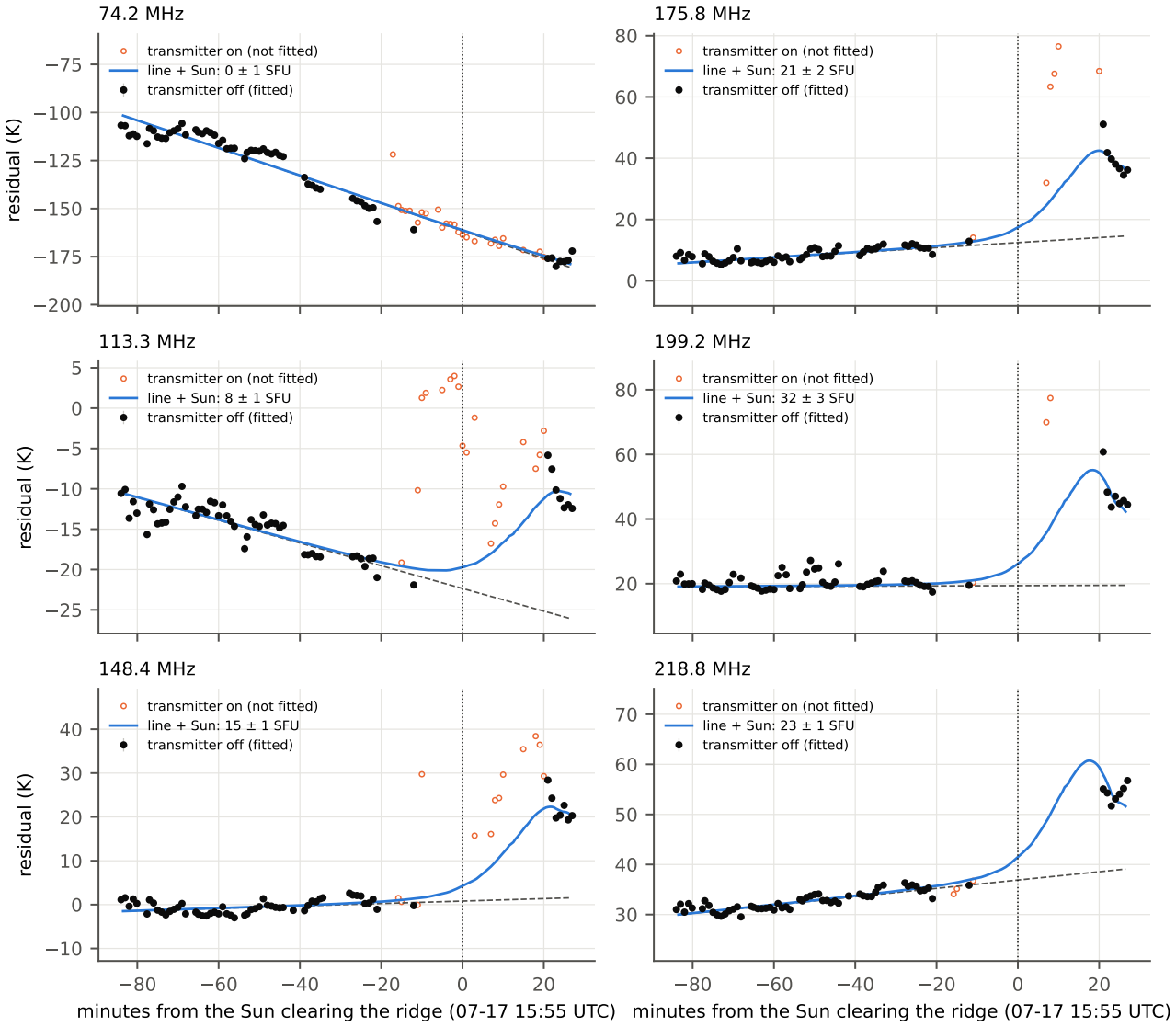



Figure 4: Out-of-sample residuals of the night fit around the ridge crossing (dotted line), at six frequencies. Filled: transmitter off, fitted. Open: transmitter on, not fitted. Dashed: the fitted line alone. Blue: line plus Sun.

### 6.3 Result

- **Spectrum.** At 31 frequencies between 105 and 235 MHz, $S_\odot$ runs from 7 to 33 SFU (31 positive), rising smoothly: 7±1 at 109 MHz, 15±1 at 148, 32±3 at 199 and 23±1 at 219 MHz (Fig. 6, Table 4).
- **Null.** At 0° sunrise the same fit gives −6 to 1 SFU over the same band. No step appears when the Sun rises behind the ridge.
- **Timing.** $\chi^2$ is smallest at $\tau = -3$ min, and within 5 of that minimum from −7 to +3 min. A shift of 20 min earlier costs $\Delta\chi^2/(\chi^2/\mathrm{dof}) = 67$ on average.
- **Kelvin scale.** If the factor $a \approx 2$ lies in the kelvin scale or the beam, it scales $S_\odot$ too. $S_\odot/a$ is 3, 8 and 14 SFU at 109, 148 and 199 MHz.
- **Below 90 MHz** the result is not usable. The residual drifts by tens of kelvin across the window there, and the null fails (grey and orange points below 90 MHz in Fig. 6).

**The spectra themselves.** Figure 5 sets aside the local fit and compares spectra. It averages the calibrated spectrum (weighted means) over two windows of transmitter-off bins. "Before" is −40 to −15 min from the ridge crossing, with the Sun behind the ridge (15 bins). "After" is the 7 bins with the Sun above it. Their difference is shown raw, and with the night model's predicted sky and ground change between the windows subtracted. That change is at most 9 K above 109 MHz, but up to 52 K below 87 MHz, where the difference is dominated by drift the model does not capture. The difference is compared with a Sun predicted with nothing fitted to these data. The prediction takes RSTN's concurrent 245 MHz flux (§6.4), extrapolated down in frequency with the MWA quiet-Sun shape (scaled by 0.85), times the Sun column averaged over each window.

- Above 109 MHz the spectrum rises by a smooth 20 K at 148 MHz and 25±1 K at 199 MHz. The flux this implies, 15 and 34 SFU, agrees with the local fit.
- The prediction is 5 and 5 K. The measured jump is 2.0–5.6 times the prediction over 31 frequencies (weighted mean 3.7). Scaled by $a$, the prediction falls short by a factor of 2.1 below 200 MHz and 1.2 above 205 MHz. The excess is flatter in frequency than the quiet-Sun shape.
- **Not a gain change.** A change in receiver gain would scale the whole spectrum, so $\Delta T/T$ would be constant. Instead it runs from 1 to 16% (coefficient of variation 0.49), while $\Delta T$ itself varies less (0.27). The jump is additive, like a new source, not multiplicative.

| MHz | $S_\odot$ (SFU) | $\chi^2$/dof | $a$ | $S_\odot/a$ | null at 12:26 |
|---|---|---|---|---|---|
| 109.4 | 6.9 ± 0.7 | 119 | 2.24 | 3.1 | -1.1 ± 0.5 |
| 125.0 | 13.3 ± 1.2 | 216 | 2.07 | 6.4 | -0.9 ± 0.6 |
| 148.4 | 14.5 ± 0.9 | 125 | 1.91 | 7.6 | -2.9 ± 1.1 |
| 175.8 | 20.9 ± 1.6 | 79 | 1.98 | 10.5 | -3.0 ± 1.7 |
| 199.2 | 32.3 ± 2.5 | 128 | 2.36 | 13.7 | -3.0 ± 1.4 |
| 218.8 | 23.5 ± 1.4 | 125 | 2.87 | 8.2 | -1.8 ± 1.2 |
| 234.4 | 25.8 ± 1.8 | 40 | 2.10 | 12.3 | 1.0 ± 1.2 |

Table 4: $S_\odot$ at selected frequencies: the local fit's $\chi^2$/dof, the GSM amplitude $a$ of `fit_v0011`, $S_\odot/a$, and the flat-horizon null at 12:26.

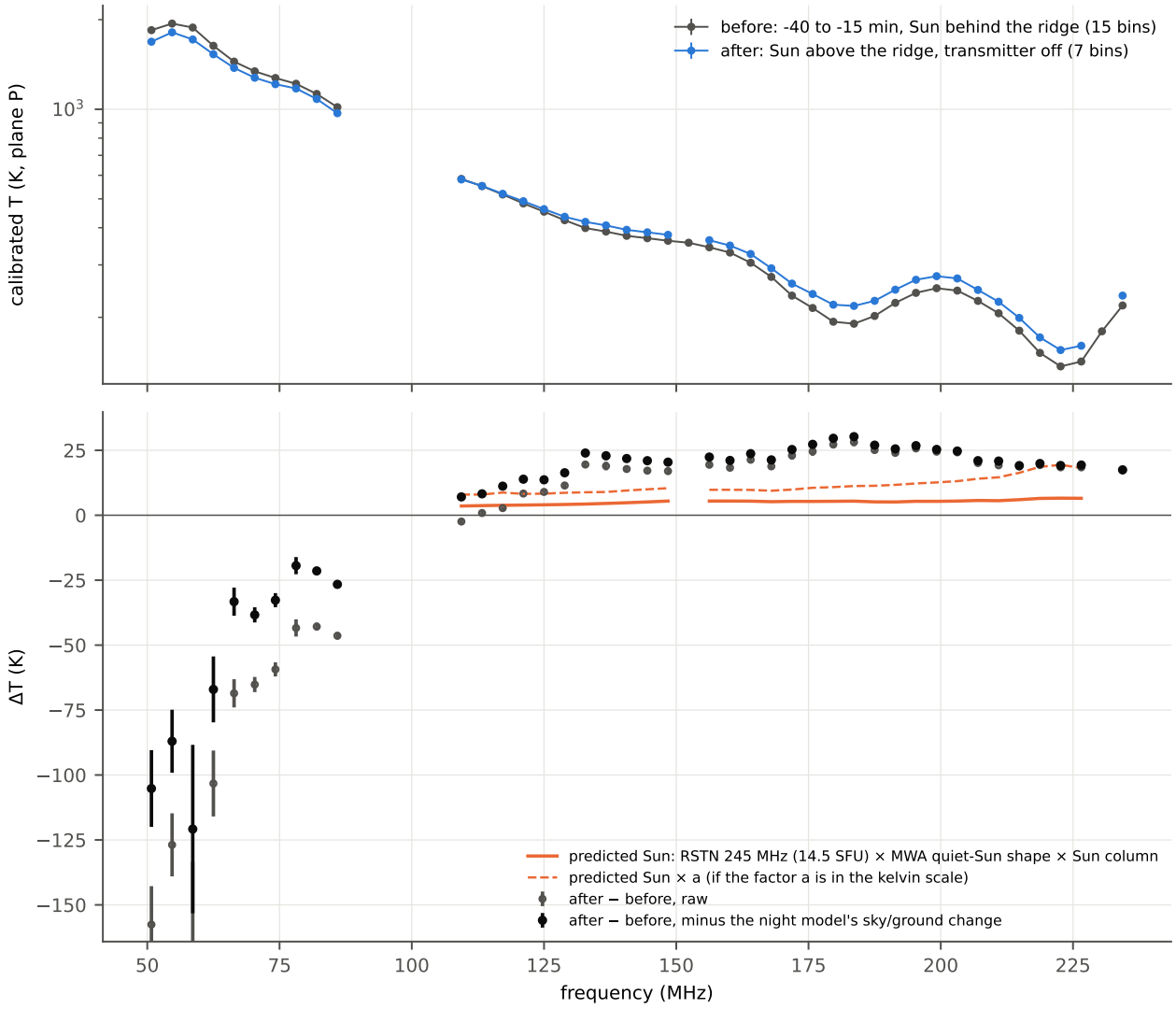



Figure 5: Top: calibrated spectra before the ridge crossing (Sun behind the ridge) and after it (Sun above, transmitter off). FM is blank. Bottom: their difference, raw (open) and with the night model's sky/ground change removed (filled). Orange: the Sun predicted from concurrent RSTN 245 MHz flux with the MWA quiet-Sun spectral shape, through the beam and the ridge; dashed: the same times the GSM amplitude $a$. Nothing in the orange curves is fitted to these data.

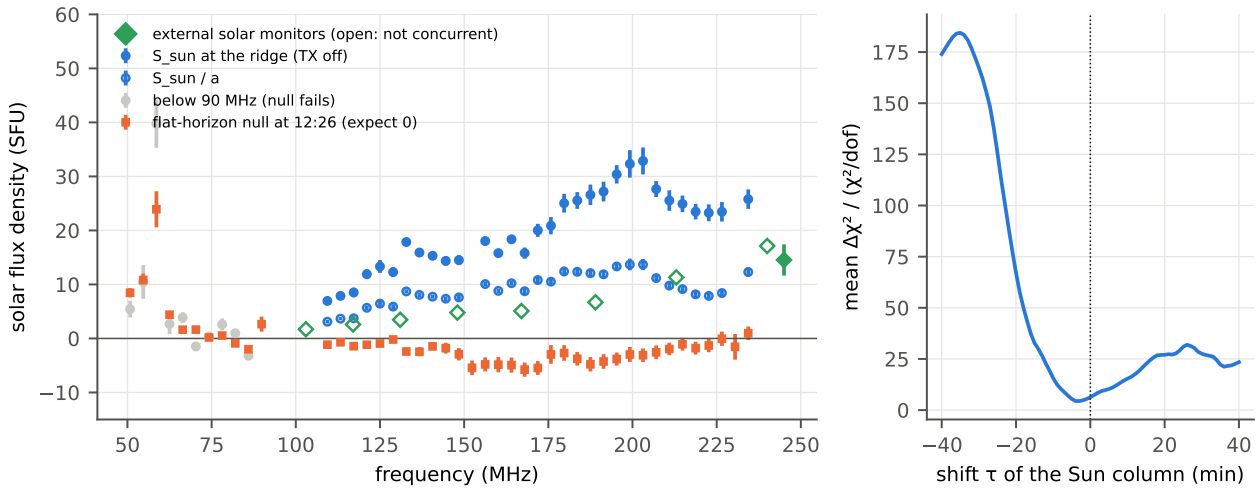



Figure 6: Left: $S_\odot$ from the transmitter-off fit at the ridge crossing (filled), divided by the GSM amplitude $a$ (open), and the flat-horizon null at 12:26 (squares). Diamonds are external solar monitors (Section 6.4). Right: mean $\Delta\chi^2$ over frequencies against a shift of the Sun column; $\tau > 0$ is later than the DEM predicts.

## 6.4 Concurrent solar monitoring

The USAF Radio Solar Telescope Network (RSTN) measures the whole-Sun flux at 245, 410 and 610 MHz every second. Two stations recorded continuously through 15:54–16:22 UTC on 07-17. At 245 MHz Sagamore Hill reads 14.9 SFU (rms 0.8, 99th percentile 18) and San Vito 14.1 SFU (rms 0.7, 99th percentile 17), over 1651 and 1650 one-second samples. Palehua's reading is below 5 SFU for 52% of the window, as if not yet tracking the Sun; over the transmitter-off bins with the Sun up it reads 25 SFU. The stations' noon values for 07-17 at 245 MHz are Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU. The stations disagree by more than the quoted 2–3 SFU; we take the two continuous stations' mean, 14.5 SFU, as the concurrent reference at 245 MHz, and note that Palehua reads higher.

The Sun was quiet: SWPC lists no flare or radio burst between 13:18 and 20:16 UTC, the 10.7 cm flux was 104 sfu (Penticton, 17:00 UTC), and e-CALLISTO spectrographs show no burst at 100–250 MHz in the window. Sagamore Hill's spectrograph recorded a series of type III bursts at 25–180 MHz from 12:14 to 13:18, inside the null window, when the Sun was behind the ridge. We have not looked for them in our data.

Nothing calibrated was found below 245 MHz for that day. For the spectrum's shape, the MWA's quiet-Sun measurement (Oberoi, Sharma & Rogers 2017, on 2013-09-03) gives 1.7, 3.5, 5.1, 11.3 SFU at 103, 131, 167, 213 MHz, rising with frequency as ours does, with 10–60% systematic error. At 234 MHz, the top of our band, $S_\odot$ is 26±2 SFU and $S_\odot/a$ is 12 SFU. RSTN gives 14.5 SFU at 245 MHz, a ratio of 1.8 ($S_\odot$) or 0.8 ($S_\odot/a$). Between 105 and 200 MHz, $S_\odot$ is 3.0–5.0 times the MWA quiet-Sun curve, and $S_\odot/a$ is 1.4–2.4 times it. The excess thus agrees with the concurrent Sun to within the unresolved factor $a \approx 2$. That agreement is not a test that separates the Sun from another effect of similar size at sunrise.

| source | when | what it says |
|---|---|---|
| RSTN Sagamore Hill, 245 MHz | 15:55–16:22 | 14.9 SFU mean, rms 0.8 |
| RSTN San Vito, 245 MHz | 15:55–16:22 | 14.1 SFU mean, rms 0.7 |
| RSTN Palehua, 245 MHz | 16:14–16:22 | 25.0 SFU mean; below 5 SFU (as if not tracking) for 52% of 15:55–16:22 |
| RSTN noon values, 245 MHz | 07-17 local noon | Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU |
| Penticton F10.7 | 17:00 | 103.9 sfu observed |
| SWPC event list | all day | 0007-0008 PAL RBR 245 820; 0150-0150 PAL RBR 245 260; 1214-1318 SAG RSP 025-180 VI/2; 2016-2053 SAG RSP 025-076 VI/2; 2025-2114 G19 XRA 1-8A B6.5 1.8E-03; 2030-2034 PAL RSP 025-075 III/2; 2214-2251 PAL RSP 025-055 VI/2; 2325-2325 PAL RSP 025-037 III/2 |
| e-CALLISTO (6 stations) | 15:45–16:30 | no burst at 100–250 MHz (uncalibrated, visual) |
| MWA quiet Sun (Oberoi et al. 2017) | 2013-09-03 | 103 MHz 1.71, 117 MHz 2.6, 131 MHz 3.46, 148 MHz 4.79, 167 MHz 5.08, 189 MHz 6.69, 213 MHz 11.3, 240 MHz 17.12 SFU |

Table 5: External solar radio measurements for 2026-07-17 (`inputs/solar_reference.json`, with URLs).

## 6.5 Solar radio bursts in the raw data

A radio burst is a property of the Sun, recorded independently by other observatories. It is the most direct test of whether box-air receives solar emission. Type III bursts drift from high to low frequency within seconds and last tens of seconds; at 0.54 s per integration, box-air resolves them.

**The reference list.** `curation/solar_bursts_v0001.json` lists the 32 bursts that e-CALLISTO spectrographs recorded at two or more stations while the Sun was above the Marjum ridge (Table 3). It starts from e-CALLISTO's curated burst list for July 2026, a detector that requires three stations, with times to the minute. Spectrograms from 11 stations in daylight were then inspected to time each burst to about a second and to add bursts the list missed. All are type III or groups of type III. The stations cover 5–93 MHz, apart from one noisy one at 46–226 MHz, so the list cannot say whether a burst reached our higher frequencies.

**Method.** For each burst, `sun_bursts.py` reads raw box-air antenna rows from 90 s before to 90 s after the burst, with *no* RFI flags, since the flags mark positive excesses, which is what a burst is. It divides each channel by its median over that cutout. It then measures the excess in four bands during the burst, in robust standard deviations of the same band away from it, and finds when each 10 MHz sub-band peaks. Box-air antenna data exist at 12 of the 32 bursts; the rest fall in data gaps or calibration cycles (Table 6).

**What the raw data show.** Figure 7 shows three bursts as raw dynamic spectra, each channel divided by its median.

- **07-15 18:33 (30 m).** A narrow ridge of emission appears inside the e-CALLISTO interval and drifts from the 70 MHz range down to the bottom of our band. The 10 MHz sub-bands peak in sequence, at −8 MHz/s. In the light curves, 45–65 MHz rises by 99% of the raw power ($115\sigma$), while 170–235 MHz does not move ($1\sigma$).
- **07-15 18:23 (30 m).** Two components about 25 s apart, both low-frequency only, $119\sigma$ at 45–65 MHz. The brighter, later one drifts at −9 MHz/s.
- **07-17 20:33 (91 m, during the raster).** The brightest burst of the period (e-CALLISTO's "IIIG/3", a group). Several components sweep down to the bottom of our band. Sub-bands up to 120 MHz peak above $8\sigma$ and up to 220 MHz above $5\sigma$, so the group reaches above the reference stations' bands. The raw power at 45–65 MHz rises by 120%. The raster's own modulation is visible behind it.

**Test.** A random time in daytime is often near one of the local spikes of §6.6, so a big excess alone is not a detection. The bursts e-CALLISTO saw are bright at 45–85 MHz and fainter above, while the local spikes are

broadband. The test therefore counts times that are bright at 45–65 MHz *and* quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$), at the bursts and at 250 random times in the same Sun-up windows, at least 5 min from any listed burst (Fig. 8). 5.2% of the random times pass, against 5 of the 12 bursts with data; the chance of that many by accident is $2.2 \times 10^{-4}$. This test could have failed.

**What this shows.** Box-air records solar radio bursts. They come at the times independent stations recorded them, drift from high to low frequency as type III bursts do, and have the low-frequency-weighted spectrum those stations saw. That is a physical identification of solar emission in our data, not a coincidence in timing. It says nothing about the quiet Sun's flux, since the e-CALLISTO spectrograms are not calibrated.

| burst start (UTC) | dur. (s) | e-C MHz | sta. | box-air excess (robust $\sigma$) 45–65 | 65–85 | 110–170 | 170–235 | match |
|---|---|---|---|---|---|---|---|---|
| 07-15 18:21:37 | 14 | 20–85 | 4 | 6 | 3 | 16 | 1 | no |
| 07-15 18:23:19 | 43 | 20–85 | 5 | 119 | 36 | 10 | 2 | yes |
| 07-15 18:33:02 | 19 | 10–85 | 6 | 115 | 4 | 1 | 1 | yes |
| 07-15 18:47:01 | 12 | 10–45 | 2 | no box-air antenna data | | | | – |
| 07-15 22:45:10 | 5 | 10–45 | 2 | no box-air antenna data | | | | – |
| 07-15 22:59:08 | 8 | 10–40 | 2 | no box-air antenna data | | | | – |
| 07-15 23:49:00 | 10 | 15–50 | 2 | 11513 | 1796 | 3046 | 2184 | no |
| 07-16 16:23:55 | 19 | 20–85 | 6 | 19 | 1 | 2 | 4 | yes |
| 07-16 16:50:02 | 11 | 20–85 | 6 | no box-air antenna data | | | | – |
| 07-16 16:54:58 | 13 | 20–90 | 7 | 61 | 11 | 72 | 19 | no |
| 07-16 17:16:06 | 26 | 20–85 | 6 | no box-air antenna data | | | | – |
| 07-16 17:25:35 | 5 | 46–85 | 3 | no box-air antenna data | | | | – |
| 07-16 17:26:22 | 10 | 20–85 | 6 | no box-air antenna data | | | | – |
| 07-16 17:57:21 | 50 | 46–80 | 3 | no box-air antenna data | | | | – |
| 07-16 18:33:13 | 31 | 20–85 | 6 | no box-air antenna data | | | | – |
| 07-16 18:48:05 | 11 | 20–65 | 3 | no box-air antenna data | | | | – |
| 07-16 19:17:07 | 16 | 20–85 | 5 | no box-air antenna data | | | | – |
| 07-16 21:36:13 | 8 | 20–55 | 2 | no box-air antenna data | | | | – |
| 07-16 22:05:44 | 10 | 15–45 | 2 | no box-air antenna data | | | | – |
| 07-16 22:25:10 | 9 | 15–50 | 2 | no box-air antenna data | | | | – |
| 07-16 22:28:10 | 16 | 15–60 | 3 | no box-air antenna data | | | | – |
| 07-16 22:32:42 | 9 | 55–85 | 3 | no box-air antenna data | | | | – |
| 07-16 23:58:59 | 34 | 20–75 | 2 | 17 | 9 | 16 | 20 | no |
| 07-17 20:16:26 | 14 | 30–85 | 4 | 44 | 9 | 3 | 0 | yes |
| 07-17 20:30:45 | 46 | 10–60 | 5 | 8 | 2 | 2 | 4 | no |
| 07-17 20:32:59 | 37 | 10–90 | 6 | 35 | 10 | 5 | 4 | yes |
| 07-17 20:34:14 | 23 | 10–85 | 5 | 5 | 2 | 0 | 1 | no |
| 07-17 21:21:22 | 99 | 10–60 | 4 | 4 | 4 | 2 | 2 | no |
| 07-17 22:26:42 | 49 | 15–45 | 2 | no box-air antenna data | | | | – |
| 07-17 22:46:54 | 19 | 20–60 | 4 | no box-air antenna data | | | | – |
| 07-17 22:51:02 | 21 | 15–40 | 2 | no box-air antenna data | | | | – |
| 07-17 23:25:37 | 17 | 15–45 | 2 | no box-air antenna data | | | | – |

Table 6: The 32 reference bursts (`curation/solar_bursts_v0001.json`): duration and frequency range at the e-CALLISTO stations, number of stations, and box-air's excess in four bands. "match": bright at 45–65 MHz and quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$).

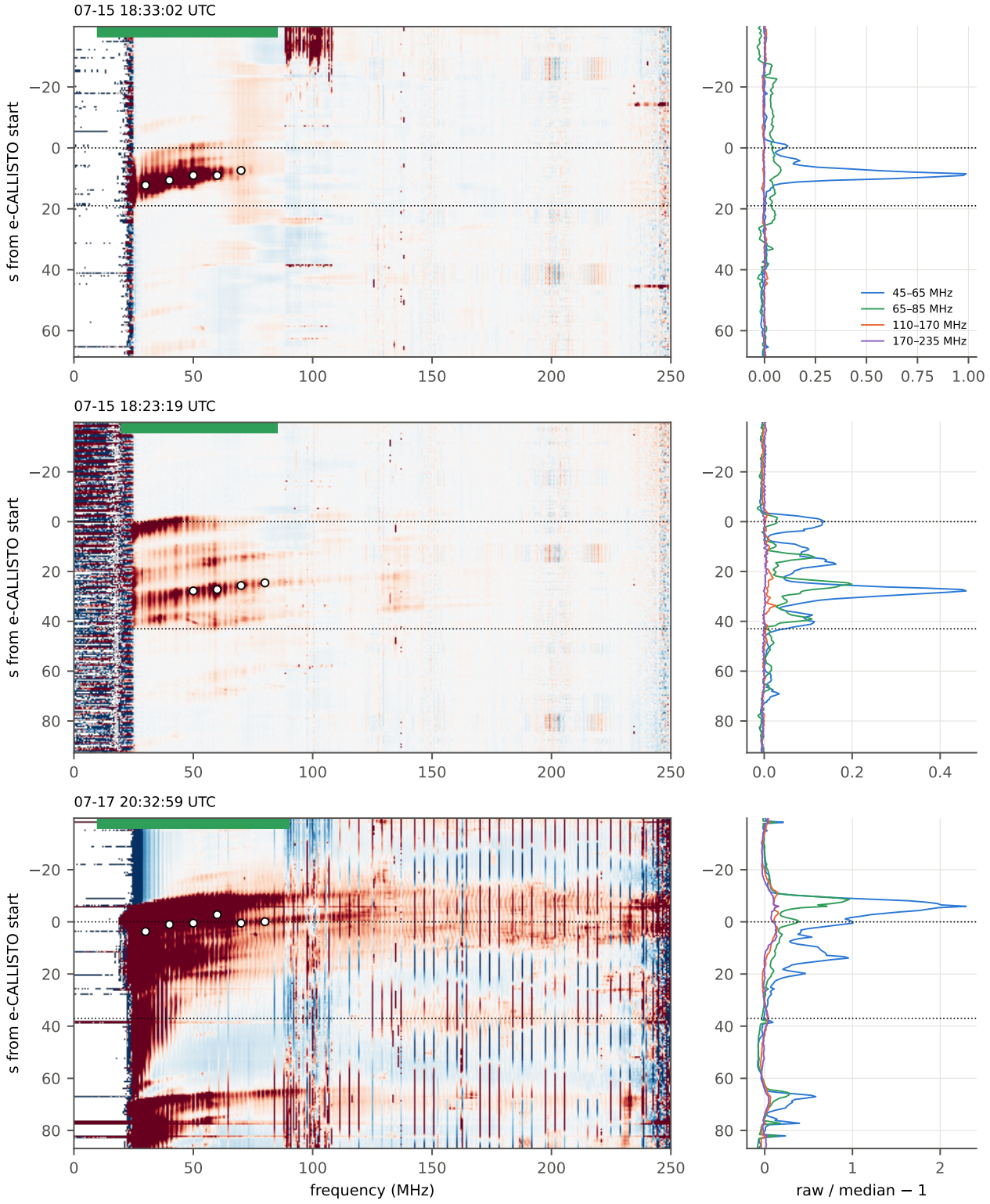



Figure 7: Three e-CALLISTO bursts in raw box-air data, time running down. Left: each channel divided by its median over the cutout, minus 1. Dotted lines are the burst's start and end at the reference stations, and the green bar along the top is their frequency range. Circles mark when each 10 MHz sub-band below 95 MHz peaks, where it is clear (above 8 robust $\sigma$) and within 8 s of the 45–65 MHz peak. Right: the same in four bands, on the same time axis.

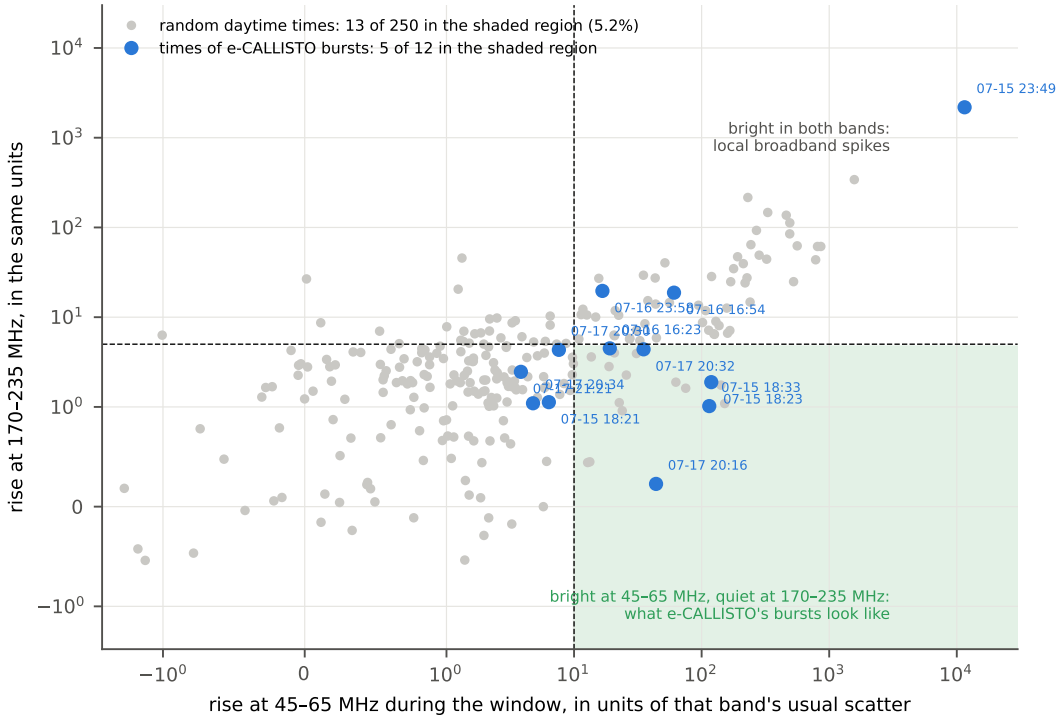



Figure 8: The burst test. Each point is one 15–40 s window of raw box-air data. Its position says how much the raw power rose during the window, at 45–65 MHz (across) and at 170–235 MHz (up), in units of that band's usual scatter. Blue: the windows of the e-CALLISTO bursts, labelled by start time. Grey: 250 random windows in the same daytime hours. The bursts e-CALLISTO recorded are bright at low frequency and faint above 170 MHz, so a real one should land in the shaded region ($z_{45–65} > 10$, $z_{170–235} < 5$). The local spikes of §6.6 rise in both bands and land at the upper right. 5 of 12 bursts fall in the shaded region, against 5.2% of the random windows. Axes are linear between −1 and 1 and logarithmic beyond.

## 6.6 Impulsive events in the raw data are local

Looked at without a reference list, raw box-air is full of impulsive events: single integrations that rise at every frequency at once, about 126 times an hour in daytime (Fig. 9, as horizontal lines). They do not drift. `impulsive_scan.py` finds them in all of phase C (44 h of antenna data) and sorts them by whether the Sun is above the ridge at the antenna, up but behind the ridge, or down. Their rate follows the time of day, not the Sun: 126 per hour with the Sun visible, 123 with it behind the ridge, and 8 at night (24, 29 and 0.8 for events lasting 3 or more integrations). If they were solar, the ridge would hide them. They are local, from a daytime source at the site we have not identified, and they are why a search for bursts must start from an independent list.

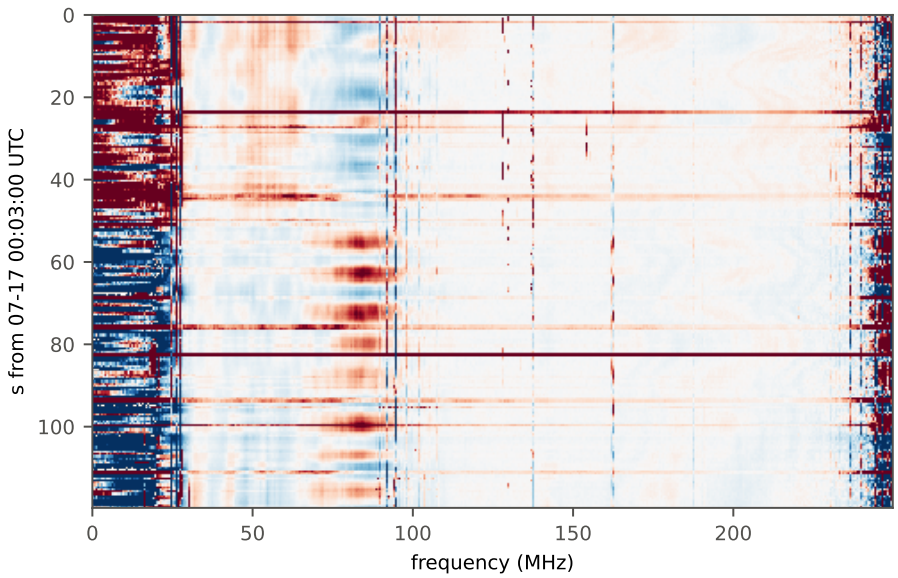



Figure 9: Two minutes of raw box-air data in daytime with the Sun behind the ridge, each channel divided by its median, time running down. The horizontal lines across all frequencies are the impulsive events.

## 6.7 Sunrise and sunset in the raw data

The calibrated data hold one terrain crossing. Phase C holds 6, the rest uncalibrated (Table 3). `sun_transitions.py` keeps the raw context of each, minute by minute, ±60 min (Figs. 10 and 11): the mean raw antenna spectrum, whether the transmitter was on, whether the antenna moved, and how many rows were antenna rows rather than calibration states.

It then fits raw power at each beam frequency as a line plus the Sun column at that crossing's pointing, over the parked stretch of data containing the crossing, without transmitter-on minutes unless the whole stretch has it on. Raw power has no kelvin scale. The fit gives $S_\odot/T_{\rm sys}$, and the 5th–95th percentile of the calibrated temperatures of 07-17 stands in for $T_{\rm sys}$. That stand-in comes from other hours and heights, so the fluxes in Table 7 are good to a factor of about 2.

**The crossings in context.** At these pointings the Sun's predicted signature in raw power is not a step but a slow bump. As the Sun sinks towards the ridge, the beam's gain toward it changes, and the ridge then cuts it off over 10–30 min. So it peaks some 20–30 min before sunset (the sunrise version is the blue curve of Fig. 4). It amounts to a few percent of the power for a Sun of a few SFU. Figures 10 (sunsets) and 11 (sunrises) show what the raw data do.

- **07-15 sunrise, 30 m**: there are no antenna rows from 2 min before the crossing to 20 min after it; no fit.
- **07-16 sunset, 30 m, transmitter off**: the quietest crossing. Raw power rises steadily, by about 10% over the two hours, at all four frequencies. The fit puts a Sun of 8–11 SFU at 199 MHz on top of that

rise, mostly from a bump 15–30 min before sunset. The bump is no larger than the unexplained wander elsewhere in the window.

- **07-16 sunrise, 30 m**: box-air's self-EMI comb (memo 006) is on until 16:51, and its lines cover much of the raw spectrum; the fitted fluxes are erratic.
- **07-17 sunset, 30 m**: the antenna is parked about 72° below horizontal, but its elevation wanders by several degrees within each minute (black ticks) until 3 min after the crossing, then settles near −65°. The abrupt step in raw power just after the crossing comes with that settling, not with the Sun. No stretch with steady pointing spans the crossing, so no fit.
- **07-17 sunrise, 87.5 m**: the transmitter is on for most of 15:36–16:14 (orange ticks), and its leakage dominates those minutes (the 10–30% excursions in the bottom panel). Calibration cycles also interrupt the antenna rows (grey). On transmitter-off minutes, including ones after the calibrated window ends at 16:22, the raw power is flat to about 1%, and the Sun term is 11–16 SFU at 199 MHz.
- **07-18 sunset, 91 m, transmitter on throughout**: fitted on channels midway between the comb teeth. The raw power wanders by several percent, and the Sun term is 7–10 SFU at 199 MHz.

**What the crossings say.** Over the 3 fitted crossings without the self-EMI and 121–219 MHz (where above $2\sigma$), the Sun term runs from 3 to 16 SFU. The concurrent prediction (RSTN at 245 MHz with the MWA quiet-Sun shape, Fig. 5) is 2–11 SFU over the same frequencies. RSTN's noon flux at 245 MHz stayed between 13 and 17 SFU from 07-15 to 07-18. So the raw data allow a Sun of the expected size. They do not detect it convincingly: the Sun term is a few-percent bump on raw power that drifts by as much for other reasons, and the kelvin stand-in is good only to a factor of 2. What they do show is that no crossing has a jump as large as the calibrated 07-17 excess implies, which is 2.0–3.2 times larger than their upper bounds (32 SFU at 199 MHz). That includes the same 07-17 crossing in raw data: with a longer stretch of transmitter-off minutes, it gives 11–16 SFU at 199 MHz.

The timing is weakly constrained. The best shifts $\tau^*$ in Table 7 range from −60 to −4 min, with broad $\chi^2$ minima; 1 of them sits at the ±60 min edge of the scan, so its shift is not bounded. The ground under the antenna goes into shadow about half an hour before the antenna at the 30 m sunsets and nearly two hours before at the 91 m one, and comes into sunlight 15–50 min after it at the sunrises ("ground" column). A signal from the lit ground, or from heating there, would therefore lead at sunset and lag at sunrise. These data cannot tell those timings apart.

| crossing | status | fit (min) | $\tau^*$ (min) | ground (min) | 121 | 148 | 199 | 219 |
|---|---|---|---|---|---|---|---|---|
| | | | | | $S_\odot$ (SFU, range from $T_{\rm sys}$ stand-in) | | | |
| rise 07-15 16:23 (30m) | no usable parked stretch | – | – | +19 | – | – | – | – |
| set 07-16 00:01 (30m) | fitted | -44 to +26 | -4 | -34 | 5–8 | 4–7 | 8–11 | 7–13 |
| rise 07-16 16:24 (30m) | fitted; self-EMI on | -42 to +10 | -16 | +19 | 1–1$^\dagger$ | 19–32 | 3–5$^\dagger$ | 6–10 |
| set 07-17 00:01 (30m) | no usable parked stretch | – | – | -34 | – | – | – | – |
| rise 07-17 15:54 (87.5m) | fitted | -44 to +38 | -4 | +50 | 3–5 | 3–4 | 11–16 | 1–2$^\dagger$ |
| set 07-18 01:21 (91m) | fitted | -44 to +10 | -60 | -114 | 8–12 | 4–7 | 7–10 | 4–7 |

Table 7: Raw-power fits at the terrain crossings (`transitions_v0003`): status, the fitted stretch and best timing shift $\tau^*$ (minutes from the crossing), when the ground under the antenna crosses ("ground", minutes), and $S_\odot$ at four frequencies (MHz) for the 5th–95th percentile of the 07-17 calibrated temperatures. $^\dagger$: below $2\sigma$.

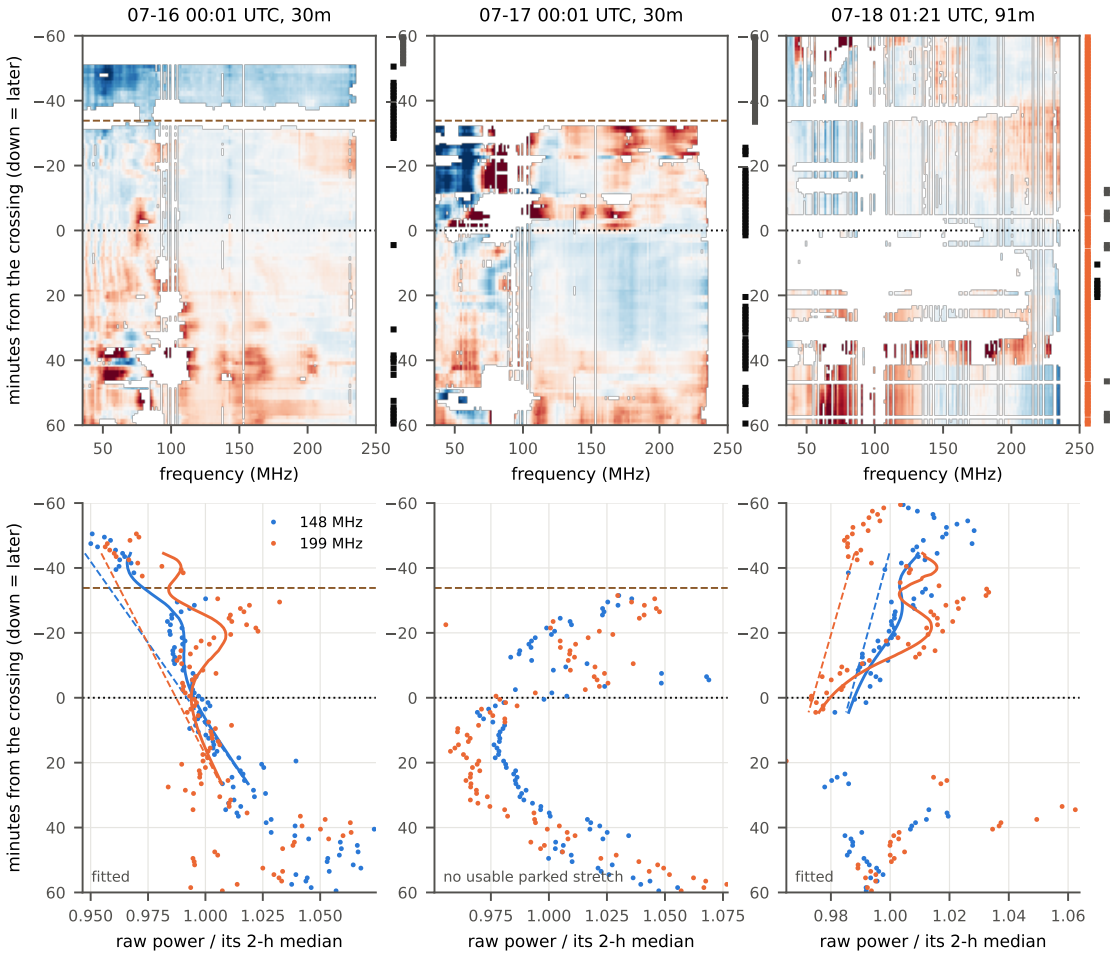



Figure 10: The three terrain sunsets in raw box-air data, one column each, time running down; the dotted line is the Sun's centre crossing the ridge at the antenna. Top: the raw spectrum, minute by minute, divided by its median over the two hours. Ticks at the right mark minutes with the transmitter on (orange), the antenna moving (black) or calibration rows (grey); the brown dashed line is when the ground under the antenna goes into shadow. Bottom: raw power at 148 and 199 MHz, divided by its median, on the same time axis. Where a fit was made, the dashed line is its baseline and the solid line baseline plus Sun. If the Sun is seen, the points follow the solid line, not the dashed one, through the half hour before the crossing.

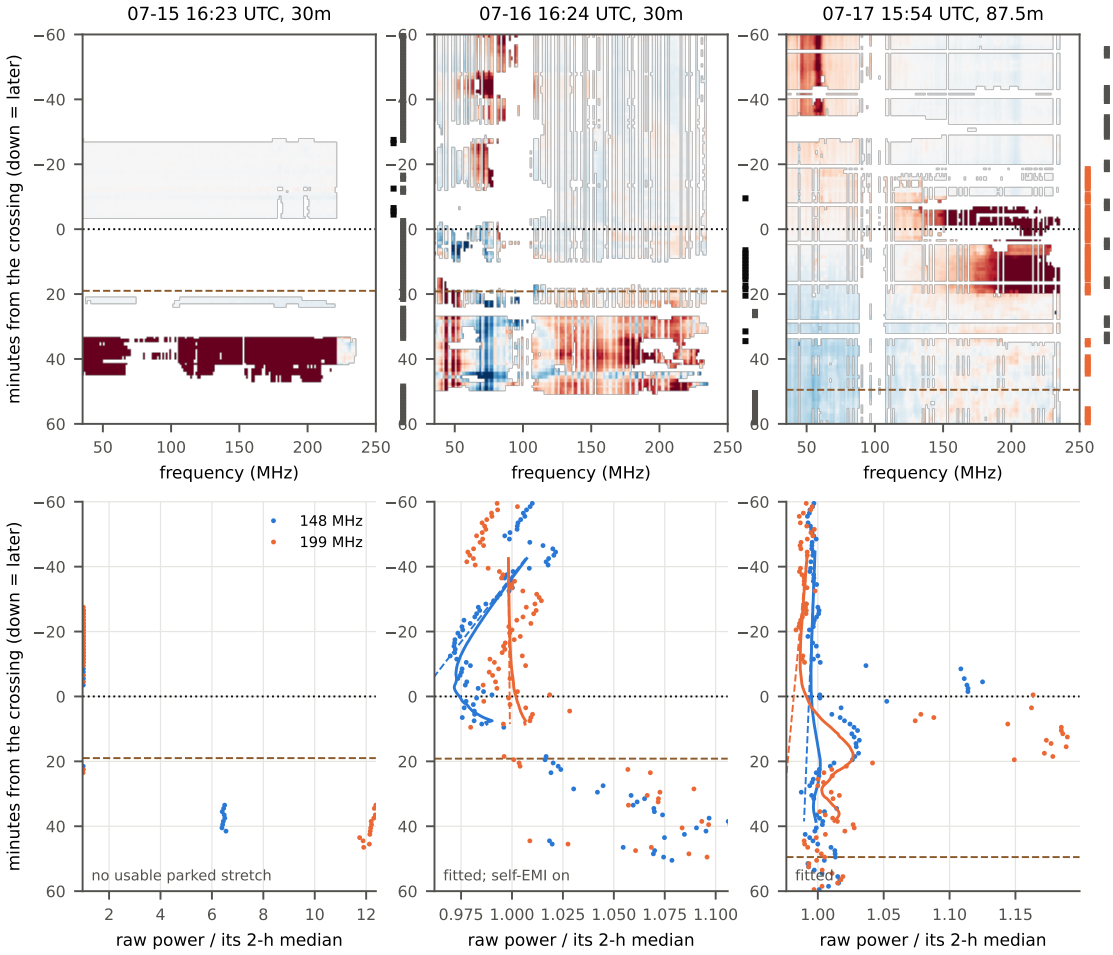



Figure 11: The three terrain sunrises, as in Fig. 10; here the brown dashed line is when the ground under the antenna comes into sunlight, and the Sun's signature follows the crossing instead of preceding it.

## 6.8 The Sun's path through the beam

Figure 12 shows the Sun column, minute by minute over phase C (`sun_profile_v0003`, 2899 minutes with pointing). Parked, it changes slowly with the Sun's motion, between the terrain crossings. Moving, it changes fast: during the 07-17 raster the beam's gain toward the Sun swings by factors of several within seconds. On 07-15 and 07-16 much of the day has no valid pointing in the table, and outside the raster the table's azimuth zero inherits potentiometer slips (memo 003). So the predicted Sun there is uncertain.

The raster offers a test with no kelvin scale. Since the receiver gain multiplies everything, regressing raw power on the sky (GSM through the beam), the ground, a constant, the transmitter's leakage, a drift and the Sun column gives the Sun's flux on GSM's brightness scale (`raster_sun.py`). The null is the same fit with fake Suns at the same altitude in 11 other azimuths. The test fails (Table 8). The real Sun beats every fake at 4 of 13 frequencies with the HFSS beam and 10 of 25 with the empirical beam, and the fitted fluxes change sign (7 and 20 negative). The residual, 1.9% of the power with HFSS, depends on direction and is larger than the Sun's modulation. Any extra direction-dependent column absorbs it.

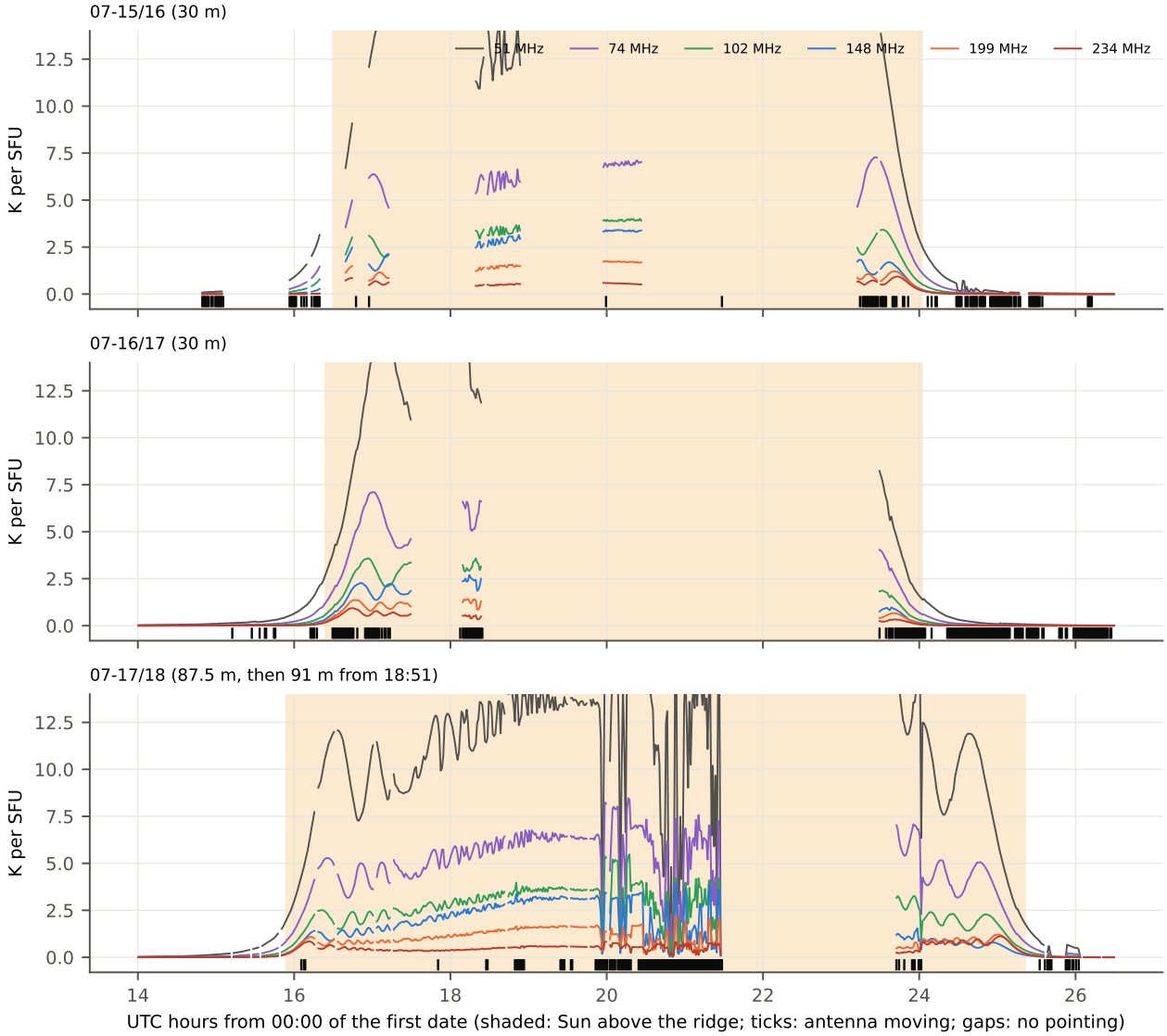



Figure 12: The Sun column (K per SFU) through box-air's beam over phase C, at six frequencies, with the knife-edge terrain visibility.

| | HFSS beam | | | | empirical beam | |
|---|---|---|---|---|---|---|
| MHz | $S_\odot$ (SFU) | fakes $\geq$ | resid. | MHz | $S_\odot$ (SFU) | fakes $\geq$ |
| 62.5 | -3 | 3/11 | 1.3% | 64.5 | -0 | 10/11 |
| 74.2 | -3 | 8/11 | 1.2% | 70.3 | -4 | 2/11 |
| 113.3 | -11 | 2/11 | 1.9% | 76.2 | -1 | 7/11 |
| 125.0 | -8 | 6/11 | 1.7% | 82.0 | -5 | 2/11 |
| 136.7 | -20 | 5/11 | 2.0% | 113.3 | -12 | 2/11 |
| 148.4 | -189 | 3/11 | 1.6% | 119.1 | -10 | 0/11 |
| 160.2 | 11 | 5/11 | 1.8% | 125.0 | -9 | 7/11 |
| 171.9 | 26 | 3/11 | 2.6% | 132.8 | -15 | 3/11 |
| 183.6 | 31 | 0/11 | 2.1% | 138.7 | -19 | 5/11 |
| 195.3 | -42 | 0/11 | 1.8% | 144.5 | -50 | 4/11 |
| 207.0 | 13 | 0/11 | 2.6% | 150.4 | -98 | 3/11 |
| 218.8 | 4013 | 0/11 | 4.3% | 156.2 | 69 | 4/11 |
| 230.5 | 12 | 1/11 | 8.8% | 162.1 | 9 | 6/11 |
| | | | | 168.0 | 39 | 1/11 |
| | | | | 173.8 | 18 | 5/11 |
| | | | | 179.7 | -323 | 0/11 |
| | | | | 185.5 | 55 | 0/11 |
| | | | | 191.4 | -193 | 0/11 |
| | | | | 197.3 | -40 | 0/11 |
| | | | | 203.1 | -17 | 0/11 |
| | | | | 209.0 | -53 | 0/11 |
| | | | | 214.8 | -42 | 0/11 |
| | | | | 220.7 | -13 | 0/11 |
| | | | | 226.6 | -21 | 0/11 |
| | | | | 232.4 | -6 | 5/11 |

Table 8: The raster test (`raster_sun_v0003`): the Sun's flux on GSM's scale, how many of the 11 fake Suns improve $\chi^2$ as much or more, and the residual rms as a fraction of the power.

### 6.9 What this does and does not show

- **Box-air receives solar emission.** Solar radio bursts recorded by independent spectrographs appear in raw box-air data at the same seconds, with the type III drift and the low-frequency-weighted spectrum those stations saw (§6.5). That is a physical identification, not a coincidence in timing.
- **The raw crossings allow a quiet Sun of the expected size but do not detect it.** Across 3 uncalibrated terrain crossings on three days and three heights, the Sun term is 3–16 SFU at 121–219 MHz, against a concurrent prediction of 2–11 SFU (§6.7). It is a few-percent bump on raw power that drifts by as much for other reasons, and its kelvin stand-in is good only to a factor of 2.
- **The calibrated 07-17 excess is larger than the Sun.** At 15:54 the calibrated spectrum jumps by 3.7 times the concurrent prediction (1.5 times scaled by $a$), and the local fit gives 2.0–3.2 times the other crossings at 199 MHz. RSTN shows no brightening that day. Part of that excess is probably not the Sun. Anything that changes as the antenna and its box come into sunlight follows the same geometry, so it is not separated. The excess rests on 7 one-minute bins at the end of the calibrated window, all directly after the transmitter switches off.
- **The timing does not separate the Sun from the lit ground.** Best timing shifts range from $-60$ to $-4$ min, with broad minima, and 1 sits at the edge of the $\pm 60$ min scan.
- **The beam's sweep during the raster does not isolate the Sun** with the present sky/ground/beam model: fake Suns fit as well (§6.8).
- $S_\odot$ inherits the unresolved factor $a \approx 2$ and HFSS's gain at $\sim 50°$ from boresight.

## 7 Recovering the sky and the ground

The night fit (§5) holds one pointing at one height, so the ground temperature and the offset are degenerate there. This section tries three ways round that, each with a check that could fail.

### 7.1 Tying the frequencies together

At one pointing and height, the beam's ground fraction $f_{\mathrm{gnd}}(\nu)$ still changes with frequency. If the ground is one temperature at all frequencies (thermal emission) and the receiver offset is a smooth function of frequency,

the pedestal $f_{\rm gnd}(\nu)T_{\rm gnd} + T_{\rm off}(\nu)$ could separate them. `joint_fit.py` reduces each frequency of the night fit to a likelihood for $(T_{\rm gnd}, T_{\rm off})$, marginalizing the sky and the GSM amplitude. It then fits one $T_{\rm gnd}$ and a Legendre polynomial in $\log\nu$ for the offset over 105–235 MHz. If the data fixed $T_{\rm gnd}$, it would not move with the polynomial's order, nor between the halves of the band. It does: 323 to 892 K as the order goes from 1 to 6, and the two halves disagree by hundreds of kelvin (Table 9). The ground fraction's frequency dependence is too much like a smooth offset; smoothness alone does not fix $T_{\rm gnd}$.

| offset order | $T_{\rm gnd}$ (K), all | lower half | upper half |
|---|---|---|---|
| 1 | 491 ± 61 | 544 ± 236 | 1 ± 244 |
| 2 | 455 ± 133 | 476 ± 244 | -24 ± 244 |
| 3 | 323 ± 136 | 540 ± 248 | 899 ± 271 |
| 4 | 657 ± 173 | 572 ± 249 | 922 ± 271 |
| 5 | 892 ± 182 | 572 ± 249 | 927 ± 271 |
| 6 | 812 ± 183 | 573 ± 249 | 931 ± 271 |

Table 9: $T_{\rm gnd}$ from the night fit tied across frequency (`joint_v0003`), against the order of the offset's polynomial, over the whole band and its two halves.

## 7.2 The sky as calibrator on parked nights

Most nights have no kelvin scale, but the Galaxy drifts through the beam. In a stretch of fixed pointing, raw power at each frequency is fitted as gain × (GSM through the beam) + a constant + a linear drift (`sky_cal.py`). The gain is then in raw units per GSM kelvin. The constant over the gain is $T_{\rm rx} + f_{\rm gnd}T_{\rm gnd}$ in GSM kelvin, and stretches at different heights, with different $f_{\rm gnd}$, would separate the two. The receiver regime is rx-A on all three nights. A stretch enters that comparison only if the sky fits it, with a median residual under 5% of the power. Figure 13 shows the raw data with the fit.

- **07-17, 87.5 m** (calibrated): the sky fits to 0.5% rms. The gain against `tcal`'s gives $a$ =1.8–2.1 at 105–220 MHz, as the night fit found. This checks the pipeline, not where $a$ comes from.
- **07-16, 30 m**: box-air's self-EMI is on all night; with its teeth masked the sky fits to 3.8%.
- **07-15, 30 m**: the switch daemon was off, so the rows carry no switch state. Before 04:45 the raw power is a few percent of its later level. After that it stays flat at 203 MHz while the sky would change it by ±15%. Whatever the receiver saw that night, it was not following the sky, and the fit misses by 18.1%; it is not used.

That leaves two nights whose ground fractions differ by only 0.05. A 10 K difference between the nights in anything else then shifts $T_{\rm gnd}$ by about 250 K, and the fit gives −2521 to 2080 K at 120–180 MHz. Parked nights at these two heights cannot separate the ground.

## 7.3 The ground from the rotating antenna

When the antenna moves, the beam's ground fraction swings between about 0.5 and 0.75 within seconds, while the receiver gain stays put. In five moving windows (`raster_sky.py`), raw power per integration is regressed on GSM through the beam, the ground fraction, a constant, the transmitter's leakage, the Sun and a drift. The gain multiplies every column, so the ground coefficient over the GSM coefficient is $T_{\rm gnd}$ in GSM kelvin, with no kelvin calibration. Errors are jackknifed over 2-minute blocks.

**The excess in the raw data.** Figure 14 shows it in the raster's raw power, integration by integration, as the beam's ground fraction swings between 0.60 and 0.75 over each elevation sweep. Take the raw power with no RFI flags at all, and subtract every fitted term except the ground: the sky, the constant, the transmitter, the Sun and the drift. What is left at 64 and 82 MHz rises and falls with the ground fraction by 34 and 44% of the power, peak to peak. A 300 K ground would move it by a few percent. The slope of that residual against the ground fraction is the ground's brightness: 2452 and 1617 K in GSM kelvin with no flags, and 3653 and 1792 K on the RFI-flagged samples (18 and 21% of them). So at low frequency the excess is in the raw data, and the flags change its size but did not create it. At 162 MHz it is small and depends on the flags: 124 K unflagged

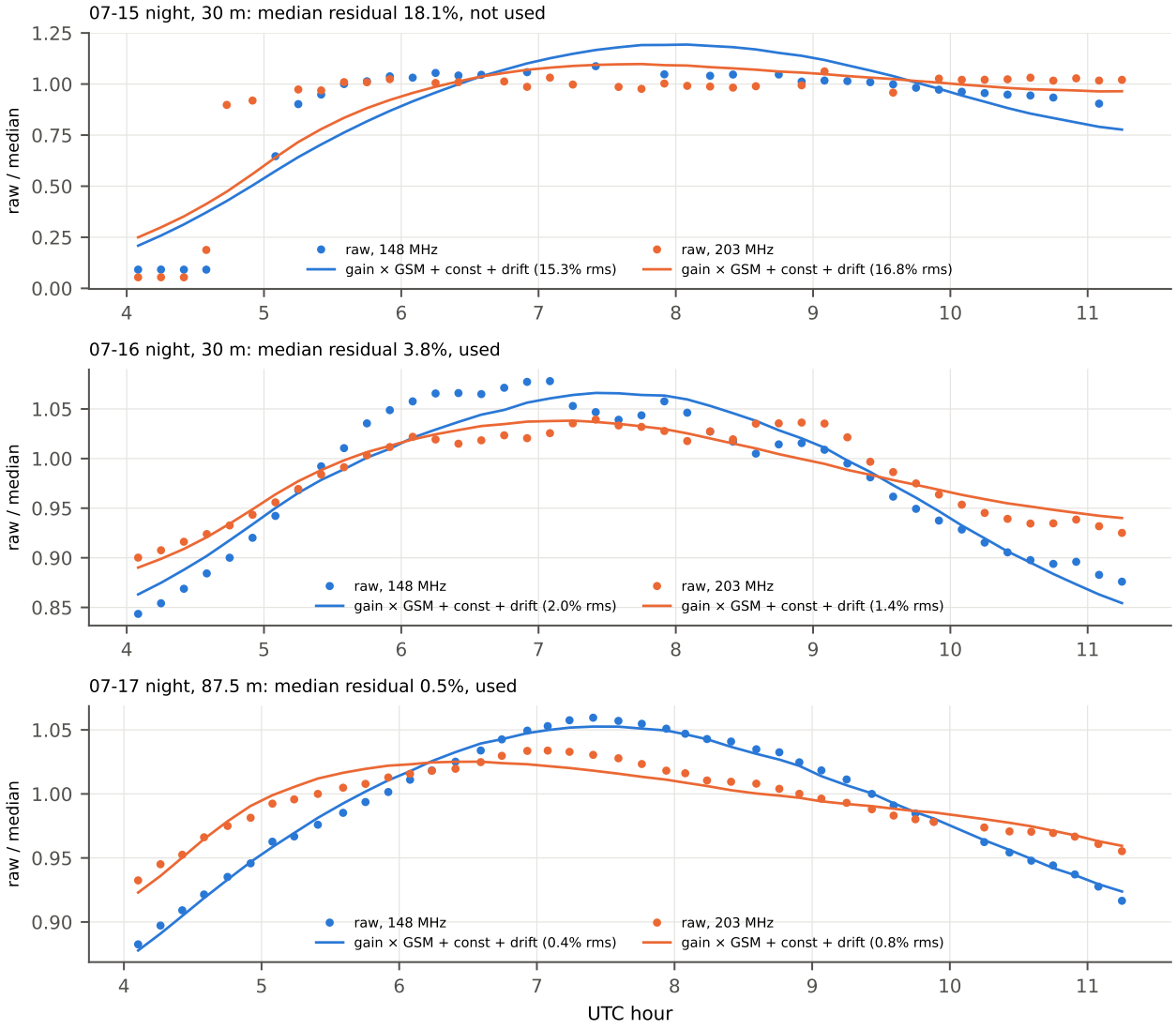



Figure 13: The three parked nights in raw data at 148 and 203 MHz, each divided by its median (points), with the fitted gain × GSM through the beam + constant + drift (lines) (`skycal_v0003`, HFSS beam). On 07-15 the raw power does not follow the sky.

and 233 K flagged (51% kept). At 113 MHz the unflagged raster is dominated by the transmitter's leakage, so only flagged values are quoted there. The flags keep fewer than a third of the raster's samples below 120 MHz, and the fraction kept depends on where the beam points. The flagged values above 100 MHz therefore carry a selection that the unflagged ones do not.

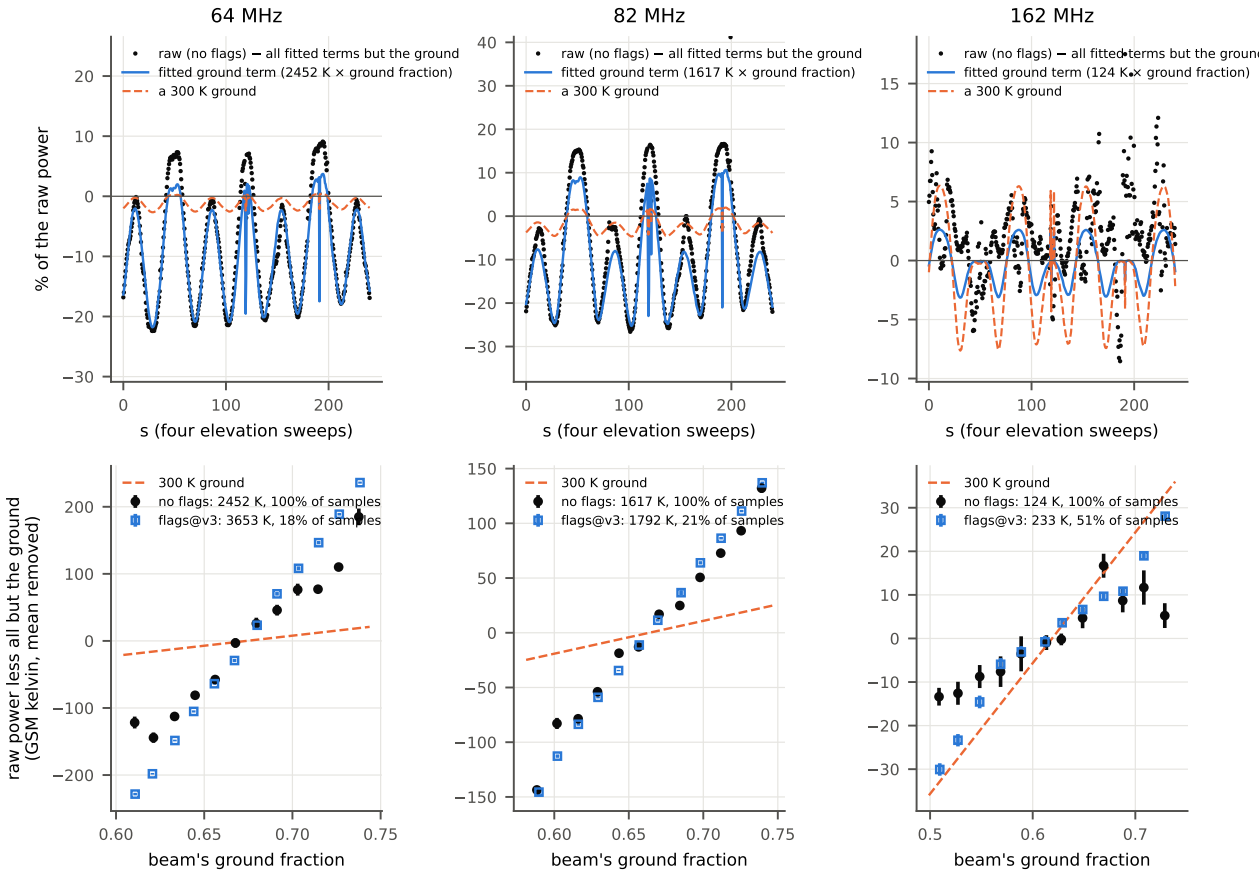



Figure 14: The ground excess in the raster's raw power (`raster_sun_v0003`, raw correlator files for the unflagged values; channels midway between the transmitter's comb teeth). Top: four elevation sweeps of raw power with no RFI flags, less every fitted term except the ground (points), against the fitted ground term (blue) and what a 300 K ground would give (orange), in percent of the power. Bottom: the same residual, binned by the beam's ground fraction and converted to GSM kelvin, with no flags (filled) and with `flags@v3` (open); the slope is the ground's brightness.

Figure 15 and Table 10 show the result in every window.

- **It repeats.** The three 91 m windows (two evenings, the transmitter and the Sun in different states) agree at 9 frequencies between 120 and 180 MHz ($\chi^2$/dof up to 6.0).
- **It is not flat in frequency.** It falls from 496 K at 125 MHz and 338 K at 139 MHz to 192–205 K at 150–175 MHz. Below 85 MHz every window gives thousands of kelvin: 1967–3653 K at 64 MHz.
- **The 30 m windows sit lower and scatter more**; the 07-16 one has the self-EMI on.

Thermal ground emission should not depend on frequency. At 150–175 MHz the values are those of soil at about 290 K with an emissivity near 0.7, but that coincidence does not identify them. What rises below 140 MHz is not the ground's temperature. It enters wherever the beam sees the ground, and it grows steeply toward low frequency like the sky.

**Not reflected sky.** Ground reflects part of the sky, which is far brighter at low frequency. A specularly reflected sky term (each below-horizon direction mirrored to the sky above, flat ground) was added in the

raster, with the ground's reflectivity $R$ free. Soil would give $R \approx 0.1$–$0.4$. At 64–82 MHz the fit gives $R = -0.13$ to $-0.04$, and the ground's emission term stays at 1840–3728 K. Specular reflection does not explain the excess. Candidates left are the beam below the horizon (both beams are extrapolated there; the raster does not sample it, memo 005) and something fixed to the platform that turns with the antenna.

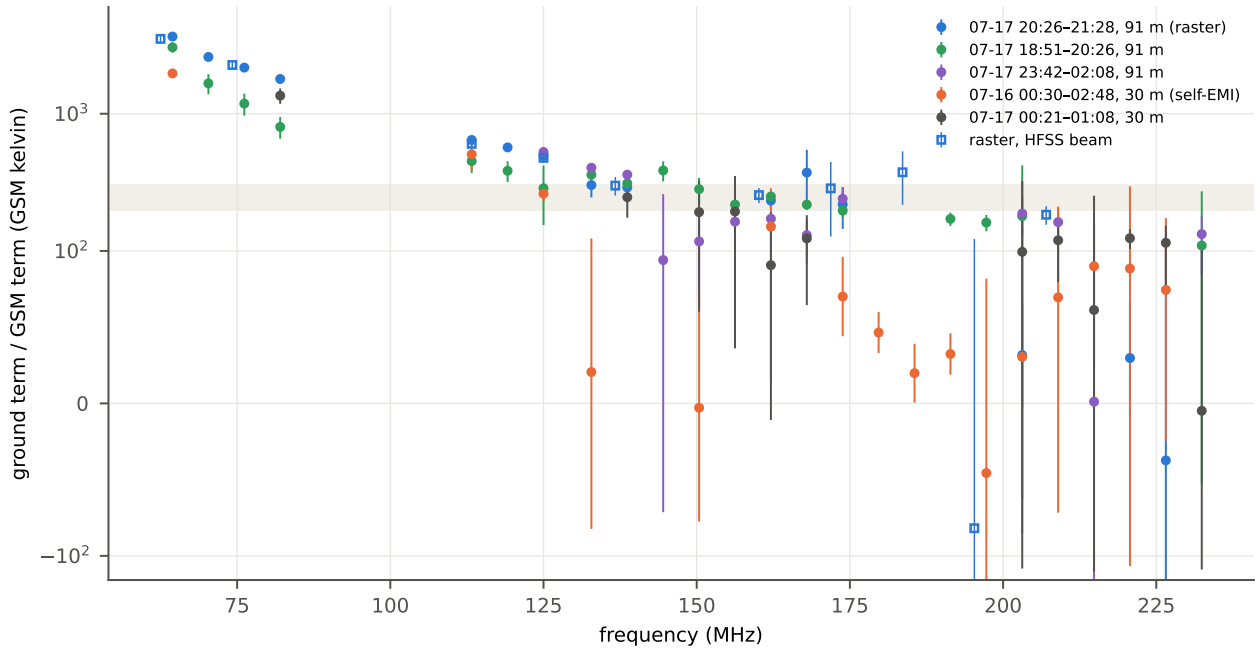



Figure 15: The ground term in each moving window, as the ratio of the ground and GSM coefficients (GSM kelvin; empirical beam unless marked; jackknife errors; points with errors above 300 K omitted). Shaded: 200–300 K. The axis is linear below 100 K and logarithmic above.

| MHz | mean of three 91 m windows (GSM K) | $\chi^2$/dof |
|---|---|---|
| 125.0 | 496 ± 31 | 2.9/2 |
| 132.8 | 382 ± 16 | 3.8/2 |
| 138.7 | 338 ± 19 | 2.0/2 |
| 144.5 | 348 ± 60 | 2.7/1 |
| 150.4 | 192 ± 40 | 4.3/1 |
| 156.2 | 194 ± 11 | 6.0/1 |
| 162.1 | 198 ± 11 | 9.9/2 |
| 168.0 | 205 ± 15 | 5.2/2 |
| 173.8 | 204 ± 19 | 0.7/2 |

Table 10: The ground term, weighted mean of the three 91 m windows, where at least two have errors under 200 K, and the windows' agreement.

## 7.4 Tracing the excess

Four explanations of the excess make different predictions, and the rotating-antenna data can test three of them.

**Where on the ground.** `ground_excess.py` repeats the regression with the ground split into regions, each with its own brightness (Table 11).

- **Horizon band against deeper ground.** Radio noise from the landscape, man-made or lightning, would arrive from just below the horizon. In the raster the band within 10° below the horizon and the deeper

ground carry similar excesses with the flags. With no flags the band is somewhat brighter, but the deep ground still carries thousands of kelvin at 64–84 MHz. The deep ground, most of the beam's ground weight, is not reached by noise from the horizon.

- **Azimuth.** Sources on the landscape would sit at particular azimuths. In the raster, eight azimuth sectors give 1811–2550 K at 76 MHz (typical error 324 K): the excess covers the ground evenly. The other windows sample too few orientations to constrain eight sectors.

**Against the sky.** A sky seen where the model expects ground would give an excess proportional to the sky's brightness. That covers sky scattered diffusely by rough ground, or a back lobe stronger than modelled. Figure 16 divides the excess (the ground term less 250 K) by GSM's whole-sky mean. In the raster the ratio is 0.84–0.99 at 64–82 MHz and falls to near zero by 150–175 MHz (within ±0.4). It also differs between windows. Its spectral index between 64 and 82 MHz is 3.3, steeper than the sky's (≈2.5). A constant scattering fraction or back-lobe gain does not reproduce this; a coupling to the sky that weakens fast with frequency would.

**A gain that follows the orientation.** If the system gain or the antenna's match changes as the antenna turns, by an amount that follows the ground fraction, the change multiplies a nearly constant system power. It then has the same shape as the ground column and cannot be told apart from it by these regressions. A gain swing as large as the ground term's swing over a sweep (Fig. 14: 34% of the power at 64 MHz, 44% at 82 MHz, 6% at 162 MHz, peak to peak) would make the whole excess. A gain change that multiplies only the sky (an extra column of ground fraction times GSM) does not absorb it: at 64–82 MHz the ground term stays at 2465–4065 K, against 1792–3653 K without it. Testing an orientation-dependent match needs the antenna's reflection coefficient at different tilts. All 19 field antenna-S11 sweeps from 07-13 on were taken upright: those with pointing (19) are within 3.2° of el = 0. So the data cannot test it.

**So far.** The excess is not specular reflection of the sky, not noise arriving from the horizon, and not concentrated in any azimuth. It does not scale with the sky as a fixed fraction. What is left is a coupling that falls steeply with frequency and follows the antenna's orientation relative to the ground: either the beam's response below the horizon, which both beam models extrapolate, or the antenna's match or gain changing as it tilts. Telling them apart needs a measurement, not more regression: S11 or a calibration cycle with the antenna tilted, or a beam measurement that samples directions below the horizon.

| window, region | 64 | 70 | 76 | 82 | 113 | 125 |
|---|---|---|---|---|---|---|
| raster, flags@v3: within 10° below horizon | 4080±251 | 2912±516 | 2260±278 | 1566±242 | 952±148 | 957±393 |
| raster, flags@v3: deeper ground | 3624±99 | 2613±170 | 2172±87 | 1785±71 | 655±49 | 523±58 |
| raster, no flags: within 10° below horizon | 5389±1054 | 2853±1029 | 3202±640 | 2424±454 | – | 282±311 |
| raster, no flags: deeper ground | 2729±773 | 1895±162 | 1816±270 | 1637±135 | – | 395±44 |
| 07-17/18 night, flags@v3: within 10° below horizon | -1604±716 | – | – | – | – | -306±222 |
| 07-17/18 night, flags@v3: deeper ground | 3554±1050 | 424±1982 | 1140±949 | 2096±710 | 782±154 | 536±24 |
| raster, 64 MHz, 8 azimuth sectors | 4124±1000, 3171±763, 5262±1465, 4625±1654, 3310±538, 5785±1129, 2010±638, 4031±598 | | | | | |
| raster, 76 MHz, 8 azimuth sectors | 2328±289, 2211±335, 2210±325, 1815±239, 2550±351, 1811±247, 2332±328, 2292±324 | | | | | |

Table 11: The ground split into regions (`ground_excess_v0003`, empirical beam; brightness in GSM kelvin, jackknife errors; "–": error above 2000 K). Top: the band within 10° below the terrain horizon and the deeper ground, at six frequencies (MHz). Bottom: the raster's eight azimuth sectors (0–45°, 45–90°, . . . , east of north).

## 7.5 Low-order sky corrections

The same regressions can add corrections to GSM: real spherical harmonics up to $l = 3$ in Galactic coordinates, whose fitted amplitudes over the GSM coefficient are corrections in GSM kelvin. Over an hour the sky turns only about 15° relative to the ground. So the same harmonics frozen to the ground (terrain, horizon or beam errors would look like that) are fitted too, and both are scored on held-out alternate 2-minute blocks (Fig. 17). The sky version beats the ground version at 14 of 25 frequencies in the raster, and in 3 of the five windows the corrections do not improve the held-out residual at most frequencies. Sky structure is not identified. With the harmonics in, $T_{\mathrm{gnd}}$ is no longer constrained either, since the lowest ones mimic the ground fraction. Telling sky from ground needs rotation at many sidereal times.

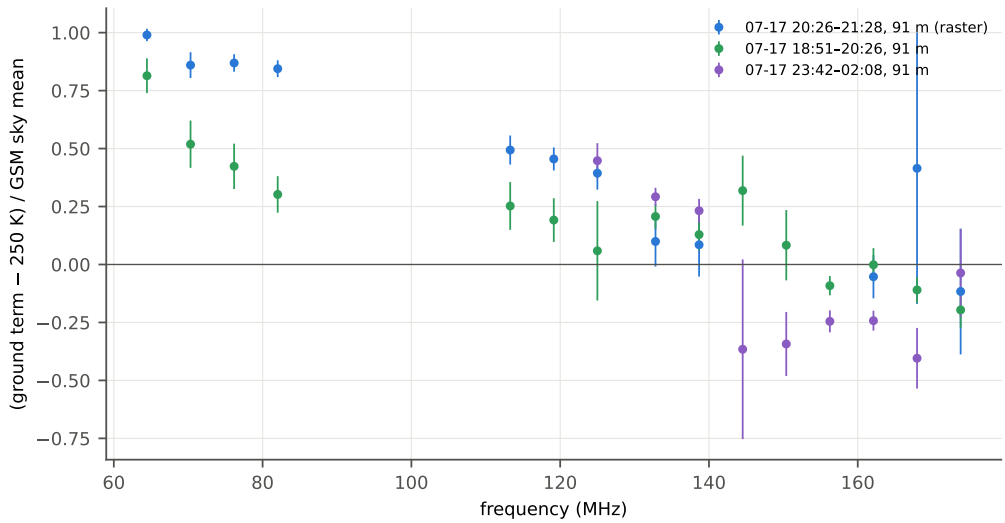



Figure 16: The ground term less a thermal 250 K, divided by GSM's whole-sky mean, in the three 91 m windows. A sky leaking in with a fixed fraction would give a constant.

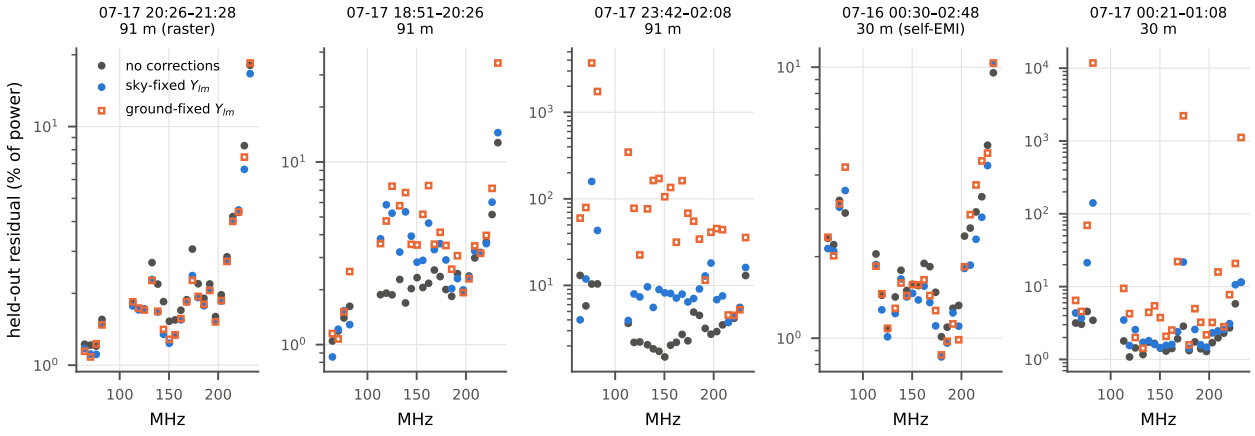



Figure 17: Held-out residual in each moving window, as a percentage of the power, without corrections to GSM (grey), with low-order sky-fixed corrections (blue) and with the same frozen to the ground (orange, open).

# 8 Open questions

- What is $a \approx 2$: the kelvin scale, the beam, or both? Memo 004's noise-source temperature and memo 005's beam bear on it directly.
- The ground temperature needs calibrated data at more than one height. Extending `tcal` to the 30 m era would give the height lever of Table 1.
- box-gnd stands upright (like box-air at el = 0) with its box on the ground. Its position there is not yet solved. Before it can give a flux scale, a box-air × box-gnd cross-correlation, or a control for the Sun or for heating, its baseline, polarization and differential horizon need substantial work. It is not used here.
- Calibrating the 30 m-era crossings (07-15 to 07-17) would turn the raw-crossing fits of §6.7 into fluxes, and test whether the 07-17 excess is the Sun.
- What makes the ground term rise below 140 MHz? It is not specular reflection, horizon noise or localized sources, and not a fixed fraction of the sky (§7.4). The beam below the horizon, or the antenna's match changing with tilt, are left. A field measurement of S11 or a calibration cycle at several tilts would separate them.
- Separating sky corrections from ground-fixed structure needs rotation at many sidereal times.
- The model is still far from the noise. Candidates are the beam (HFSS against the empirical beam), the sky prior, a time-variable ground, and the 10% GSM prior absorbing structure it cannot predict.

# Revision history

**2026-10-16** Added the effect of extending the DEM 4 km south on the nominal horizon and on the predicted temperature (Section 2, Fig. 1), and the products that follow from it: DEM v0004 (6 km more to the south), `horizon_profiles_v0004` and its coverage check. No fit or product in this memo changed.

**2026-10-07** Added the solar bursts, which identify solar emission in box-air; the raw context and raw-power fits of every terrain crossing; the local origin of the daytime impulsive events; the Sun's path through the beam and the raster test. The calibrated 07-17 excess is now read as larger than the Sun, not as the Sun. Added sky and ground recovery: the joint-frequency fit, the sky-calibrated parked nights, and the rotating-antenna regressions with their repeatable but frequency-dependent ground term. Traced the low-frequency excess: not specular reflection, horizon noise or localized sources, not a fixed fraction of the sky; the beam below the horizon or a tilt-dependent match remain, and need a field measurement.

**2026-10-06** First version: the model, the degeneracy forecast, the night fit on `tcal@v0003` and `flags@v3`, and the first look for the Sun at the 07-17 terrain sunrise.

# A Reproducing this memo

The products are regenerated, from `data-analysis/scripts/marjum-2026-07/ground_sky/` with `EIGSEP_CAMPAIGN_ROOT` set to the campaign directory, by the commands below. Every product that depends on the antenna position reads the horizon profiles `curation/horizon_profiles_v0003` (geometry release `v0004`, memo 002) and traces the Sun on the DEM mosaic `derived/dem/v0003`, and every product that rotates a beam uses the empirical beam and mount offsets of `empirical_raster_v0013`. The binned spectra, the burst cutouts and the impulsive-event scan depend on neither.

- `python bin_spectra.py v0001` and `python bin_spectra.py v0002 -bin-s 60 -min-rows 10`;
- `python fit_ground_sky.py v0009 -binned v0001 -sun cut-flat`, `v0010 -binned v0001 -sun cut-terrain`, `v0011 -binned v0002 -sun cut-flat` and `v0012 -binned v0001 -sun model`;
- `python sun_events.py $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/fit_v0011 -out-dir $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/sun_events_v0003`;
- `python sun_bursts.py v0001`, `python impulsive_scan.py v0002`, `python sun_transitions.py v0003 -sun-events v0003 -fit v0011`, `python sun_profile.py v0003`, and `python`

`raster_sun.py v0003 -beam hfss` then `-beam empirical`. The burst list they read is `curation/solar_bursts_v0001.json`.

- `python joint_fit.py v0003 -fit v0009`; `python sky_cal.py v0003 -beam hfss` and `-beam empirical`; `python raster_sky.py v0003` with `-beam hfss -label raster`, `-beam empirical -label raster -reflect`, and for the other moving windows `-beam empirical` with the `-label`, `-start`, `-stop`, `-era` and `-row-step` recorded in each output (07-16 also `-self-emi`); and `python ground_excess.py v0003` with the `-split`, `-label` and window options recorded in each output.

The commits they ran at are below; each product's `manifest.json` records them along with its inputs' checksums. This memo's figures, tables and numbers come from `python make_products.py` in this directory, then `pdflatex memo-008.tex` twice. The script checks the SHA-256 of every product file it reads.

| input | SHA-256 (first 16) |
|---|---|
| derived/ground_sky/binned_v0001/binned.npz | 6dc2f340d949dac4 |
| derived/ground_sky/binned_v0001/manifest.json | 259cddcae303f3ec |
| derived/ground_sky/binned_v0002/binned.npz | e451dd73d386c739 |
| derived/ground_sky/binned_v0002/manifest.json | bc244df8e9d42290 |
| derived/ground_sky/fit_v0009/fit.npz | c3b1935935061aee |
| derived/ground_sky/fit_v0009/manifest.json | 3fdd606f43f02bb5 |
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| derived/ground_sky/raster_sky_v0003/w0717dusk30_empirical.json | bcdc02e680612ffe |
| marjum-2026-07/curation/horizon_profiles_v0003.npz | 352415c9f6810916 |
| data-analysis/scripts/marjum-2026-07/ground_sky/summary.json | 86d8bff6f2da1566 |
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| products made with eigsep_base | 840c10b |
| products made with eigsep_data | 2b201d4, e941c51 |
| products made with eigsep_sim | d050c83 |
| eigsep (memo) commit | b3dcda5 |
| data-analysis commit (degeneracy study file) | 9e52a0a |
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